

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning

COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3,BL4,BL5)

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25

Total Marks: 50

Annexure A

Experiments

DURATION: 4 HRS (2 Sessions)

Model-Based RL Approach

1. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.
(4 Marks)
2. Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward in Model-Based Reinforcement Learning.
(4 Marks)
3. Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.
(4 Marks)
4. Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.
(4 Marks)
5. Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.

(4 Marks)

Analytic Gradient Computation

6. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.

(4 Marks)

7. Design a concept map showing how gradients are computed and propagated during policy learning.

(4Marks)

8. Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.

(4 Marks)

9. Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.

(4 Marks)

10. Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.

(4 Marks)

Sampling-Based Planning

11. Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

(4 Marks)

12. Prepare a concept map showing how sampled trajectories are generated and evaluated for planning.

(4 Marks)

13. Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy.

(4 Marks)

14. Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.

(4 Marks)

15. Draw a concept map comparing exhaustive search planning and sampling-based planning techniques.

(4 Marks)

Model-Based Data Generation

16. Draw a flowchart illustrating synthetic data generation using a learned environment model.

(4 Marks)

17. Prepare a concept map showing the relationship among environment model, generated experiences, replay buffer, and policy learning.

(4 Marks)

18. Draw a block diagram explaining model-generated experience replay in Reinforcement Learning.

(4 Marks)

19. Design a flowchart showing how simulated experiences improve sample efficiency during training.

(4 Marks)

20. Draw a concept map illustrating the advantages and limitations of model-generated training data.

(4 Marks)

Value-Equivalence Prediction and Policy Optimization

21. Draw a flowchart explaining value-equivalence prediction for estimating future rewards.

(4 Marks)

22. Prepare a concept map showing the relationship among value functions, reward prediction, policy evaluation, and policy improvement.

(4 Marks)

23. Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.

(4 Marks)

24. Design a flowchart explaining iterative policy optimization using learned environment dynamics.

(4 Marks)

25. Draw a comprehensive concept map integrating Model-Based RL, Analytic Gradient Computation, Sampling-Based Planning, Model-Based Data Generation, Value-Equivalence Prediction, and Model-Based Policy Optimization into a unified Reinforcement Learning framework.

(4 Marks)

Code of Conduct:

I _____M.Vineela(192472071)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism ; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness

1. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.

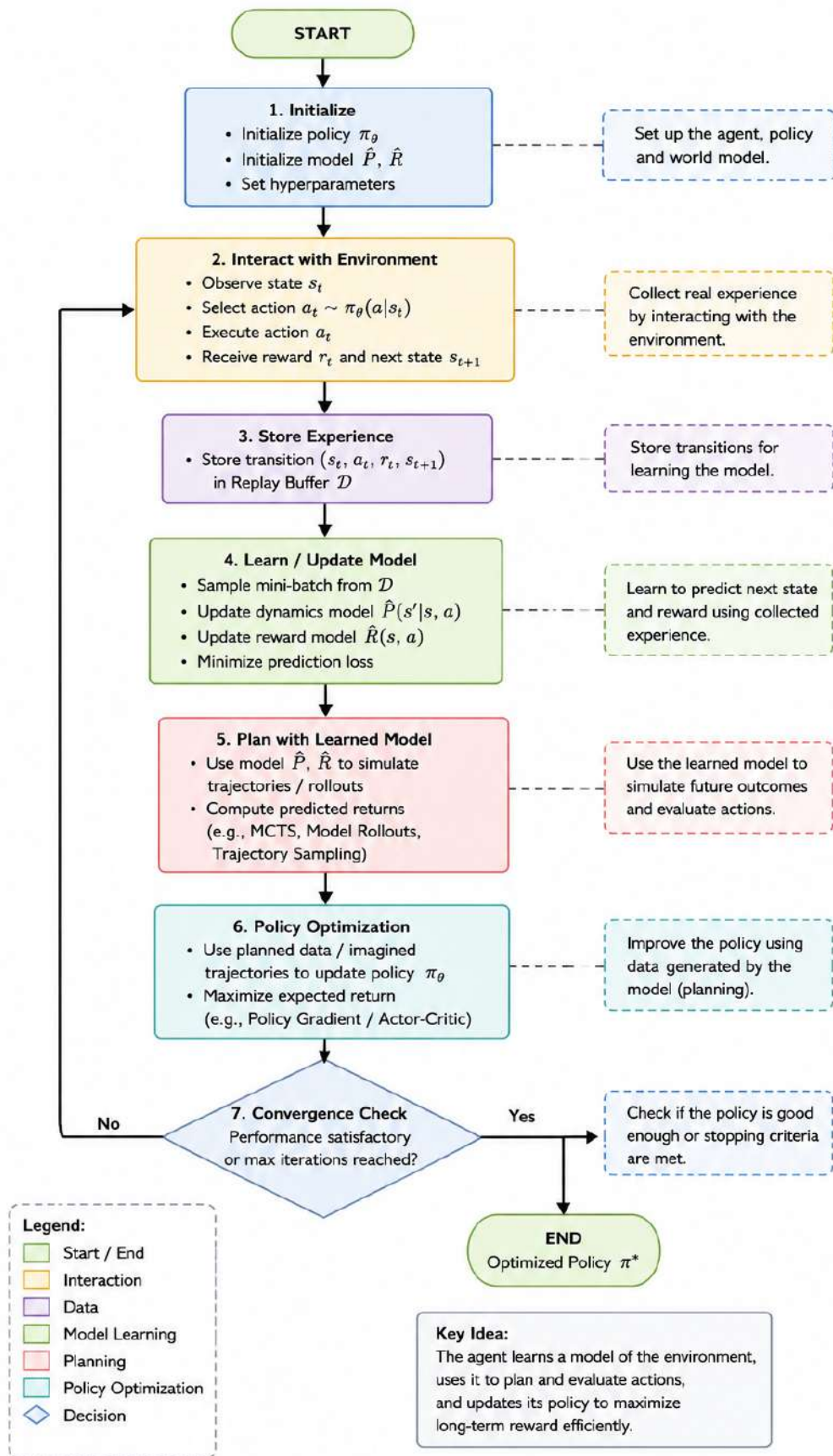
Flow:

Start → Initialize Policy & Model → Observe State → Select Action → Environment → Receive Reward & Next State → Update Environment Model → Planning → Policy Evaluation → Policy Optimization → Convergence? → Optimized Policy

1. **Initialize** – Initialize the policy and environment model.
2. **Observe State** – The agent observes the current state .
3. **Select Action** – The policy selects an action .
4. **Environment Interaction** – The environment produces the next state and reward .
5. **Update Model** – The agent learns the environment's transition and reward model.
6. **Planning** – The learned model is used to simulate future outcomes and estimate values.
7. **Policy Evaluation** – The current policy is evaluated using the estimated values.
8. **Policy Optimization** – The policy is improved based on the planning results.
9. **Convergence Check** – If the policy is not optimal, the process repeats.
10. **Optimized Policy** – When convergence is reached, the optimized policy is obtained.

Key idea: Model-Based RL learns a model of the environment, uses that model for planning, and then improves the policy.

Model-Based Reinforcement Learning (MBRL) Workflow



2. Relationship Between Environment, Model, Planning, Policy and Reward

Question:

Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward.

Concept Map

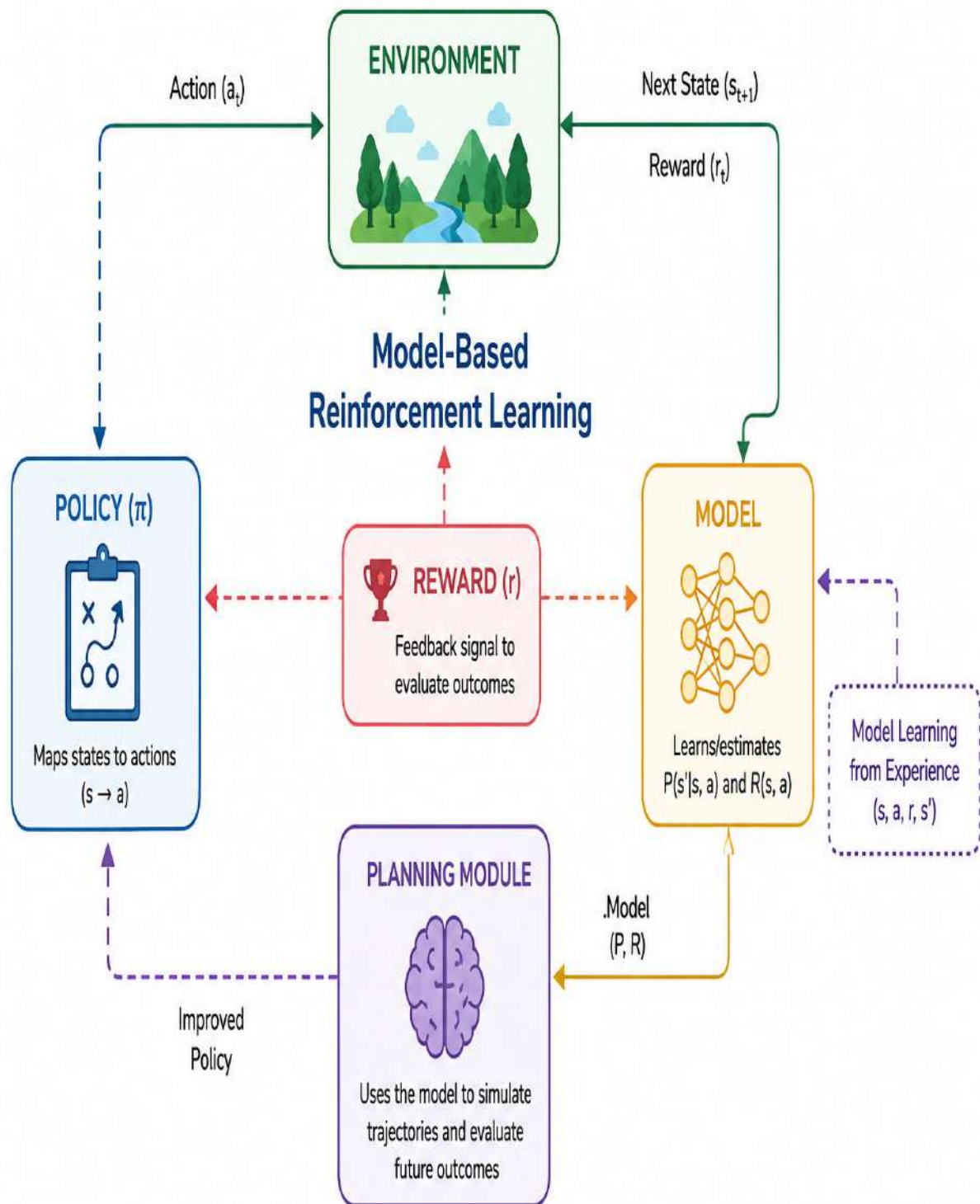
Environment → State & Reward → Model → Planning Module → Policy → Action → Environment

Simple Explanation

1. **Environment:** Provides states and rewards.
2. **Policy:** Selects actions.
3. **Reward:** Shows whether the action was good or bad.
4. **Model:** Learns how the environment behaves.
5. **Planning:** Uses the model to predict future results.
6. **Policy:** Is improved using planning results.

Key Idea

Environment provides experience → Model learns → Planning predicts → Policy improves.



3. Learning Environment Model Before Selecting Optimal Action

Question:

Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.

Flowchart

Start → Observe State → Take Action → Receive Reward & Next State → Collect Experience → Learn Environment Model → Predict Future Outcomes → Evaluate Actions → Select Best Action → Execute Action

Simple Explanation

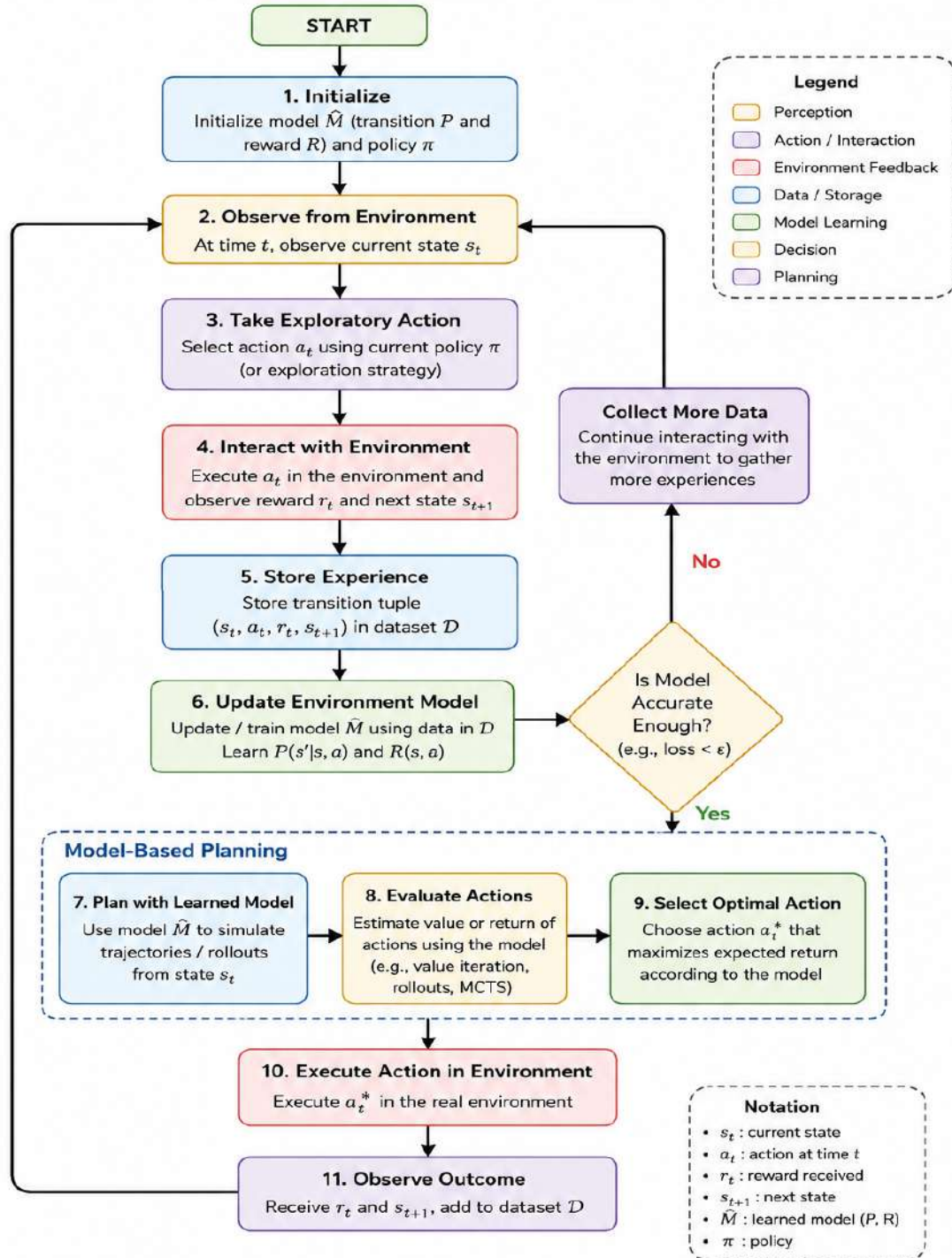
1. **Observe:** The agent observes the current state.
2. **Explore:** It takes actions and collects experience.
3. **Learn Model:** It learns environment dynamics.
4. **Predict:** The model predicts future states and rewards.
5. **Evaluate:** Different actions are compared.
6. **Select:** The best action is selected.

Key Idea

Learn the environment first → Predict outcomes → Select the best action.

MODEL-BASED REINFORCEMENT LEARNING

Learning a Model before Selecting Optimal Action



4. Architecture of a Model-Based RL System

Question:

Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.

Architecture

Environment → Model Learning → Environment Model → Planning → Policy → Action → Environment

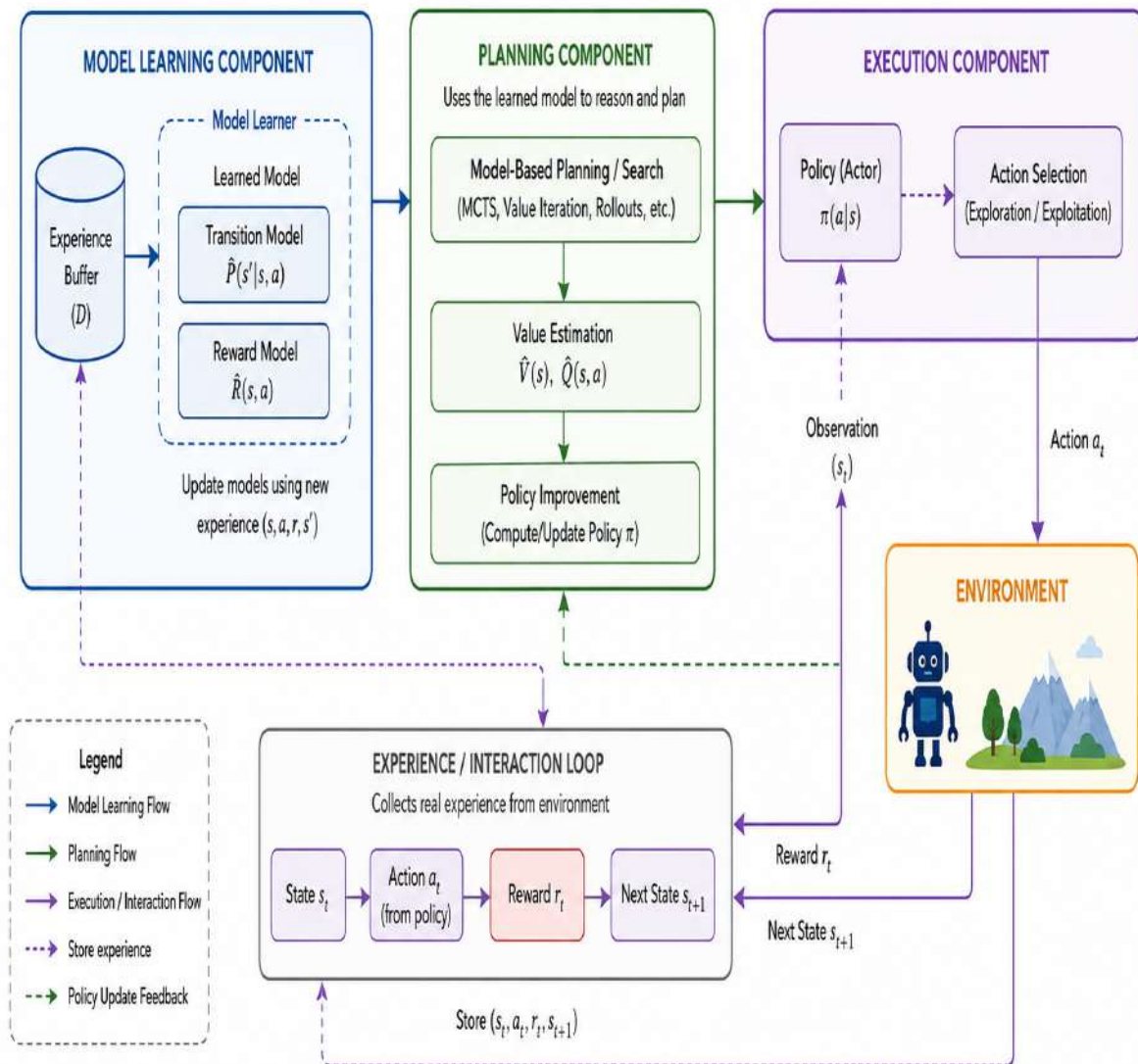
Simple Explanation

1. **Environment:** Provides real experience.
2. **Model Learning:** Learns environment behavior.
3. **Environment Model:** Represents predicted transitions and rewards.
4. **Planning:** Searches for good actions.
5. **Policy:** Selects the action.
6. **Execution:** The action is performed in the environment.

Key Idea

Model Learning + Planning + Execution = Model-Based RL Architecture

Architecture of a Model-Based Reinforcement Learning System



5. Model-Based RL vs Model-Free RL

Question: Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.

Concept Map

Reinforcement Learning → Model-Based RL / Model-Free RL

Model-Based RL → Learns Model → Uses Planning → Predicts Future

Model-Free RL → No Explicit Model → Learns Directly → Uses Experience

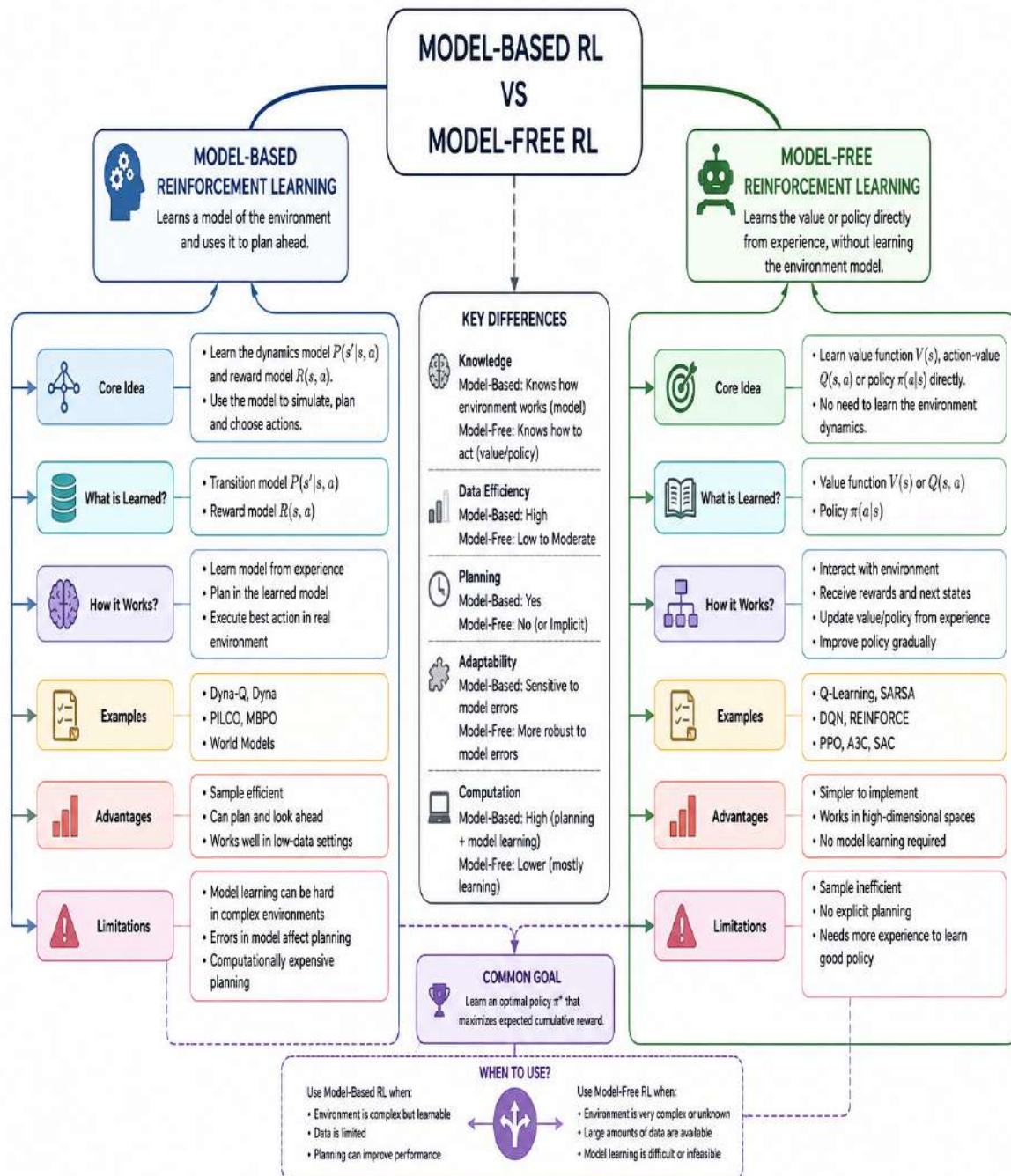
Simple Explanation

1. **Model-Based RL:** Learns how the environment works.
2. **Planning:** Uses the learned model.
3. **Model-Free RL:** Does not learn an explicit environment model.
4. **Learning:** Learns directly from experience.
5. **Goal:** Both try to maximize rewards.

Key Idea

Model-Based = Model + Planning

Model-Free = Direct Learning from Experience



Analytic Gradient Computation

6. Analytic Gradient Computation for Policy Optimization

Question: Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.

Flowchart

Start → Initialize Policy Parameters → Select Actions → Collect Rewards → Calculate Objective → Compute Gradient → Update Parameters → Evaluate Reward → Repeat → Optimized Policy

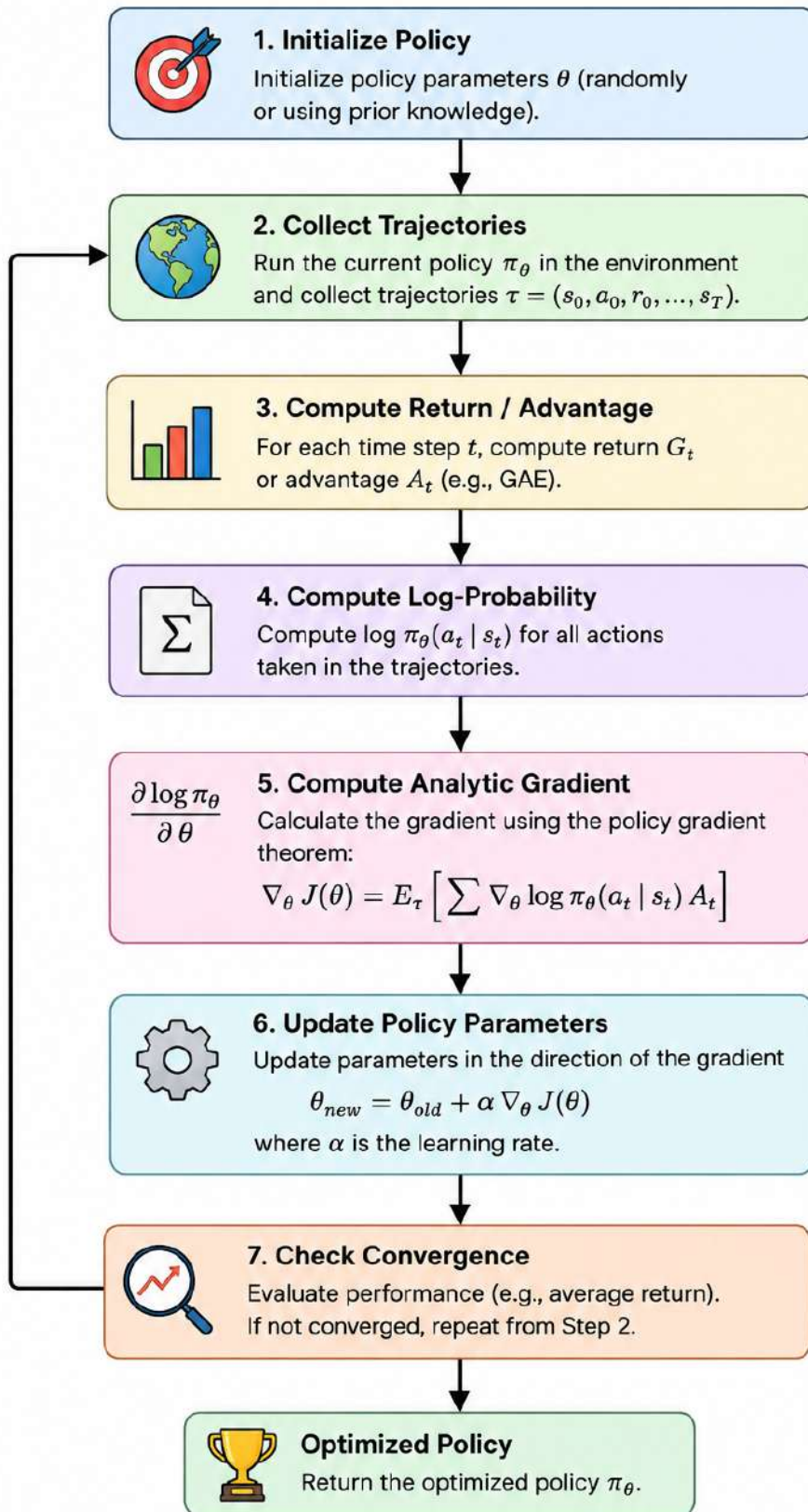
Simple Explanation

1. **Initialize:** Set policy parameters.
2. **Select Actions:** Policy chooses actions.
3. **Collect Rewards:** Environment provides rewards.
4. **Calculate Objective:** Measure policy performance.
5. **Compute Gradient:** Find the direction of improvement.
6. **Update Parameters:** Change policy parameters.
7. **Repeat:** Continue until performance improves.

Key Idea

Gradient tells the agent how to change policy parameters to increase rewards.

Analytic Gradient Computation for Policy Optimization



7. Gradients During Policy Learning

Question:

Design a concept map showing how gradients are computed and propagated during policy learning.

Concept Map

Policy Parameters → **Policy** → **Actions** → **Environment** → **Rewards** → **Objective** → **Gradient** → **Parameter Update** → **Improved Policy**

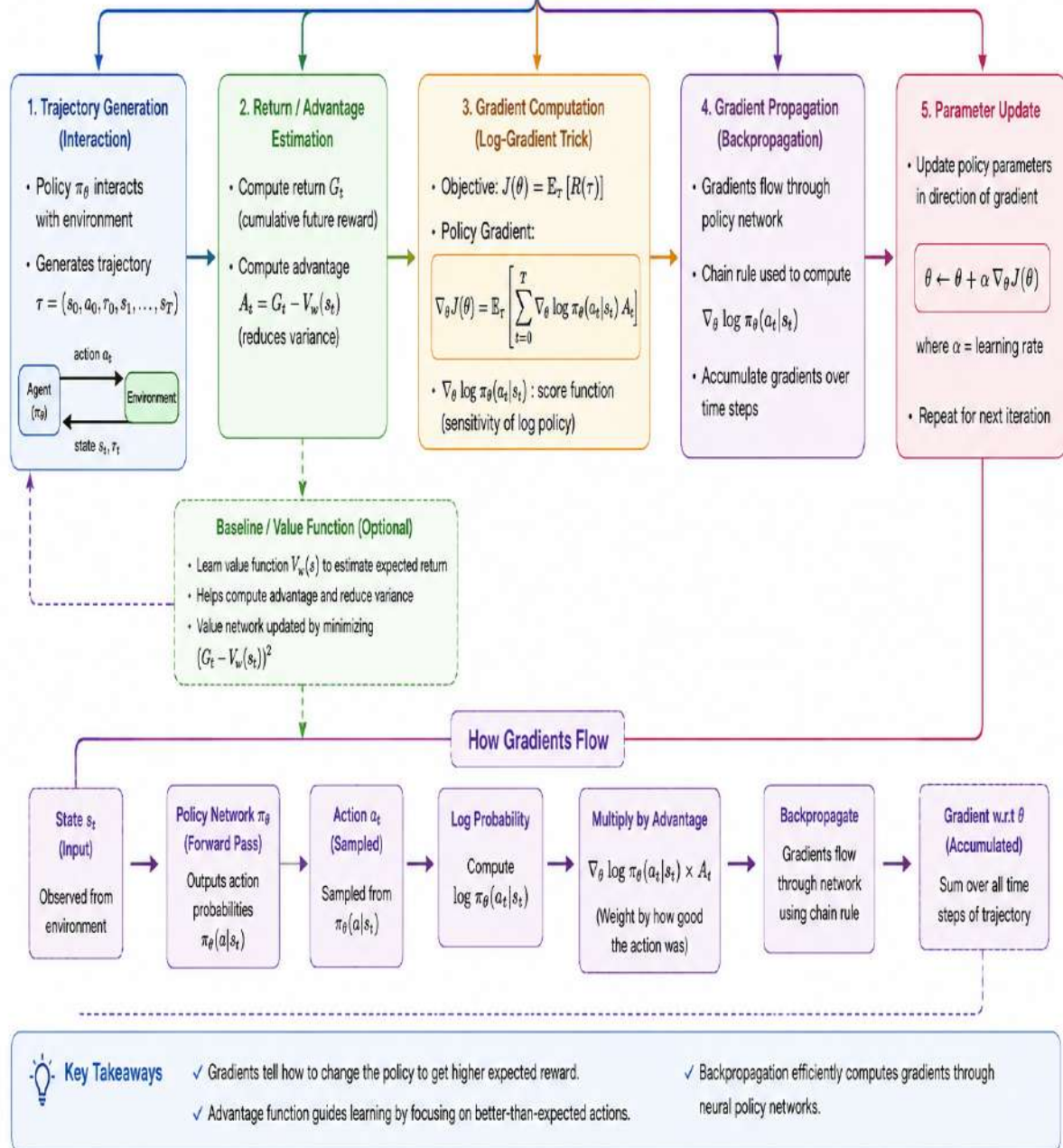
Simple Explanation

1. **Policy Parameters:** Control policy behavior.
2. **Actions:** Policy selects actions.
3. **Rewards:** Environment provides feedback.
4. **Gradient:** Shows how parameters should change.
5. **Parameter Update:** Improves the policy.

Key Idea

Rewards → **Gradients** → **Parameter Updates** → **Better Policy**

Gradients During Policy Learning



8. Gradient Computation, Parameter Updates and Reward Optimization

Question:

Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.

Block Diagram

Policy Parameters → Actions → Environment → Rewards → Gradient Computation → Parameter Update → Improved Policy → Higher Rewards

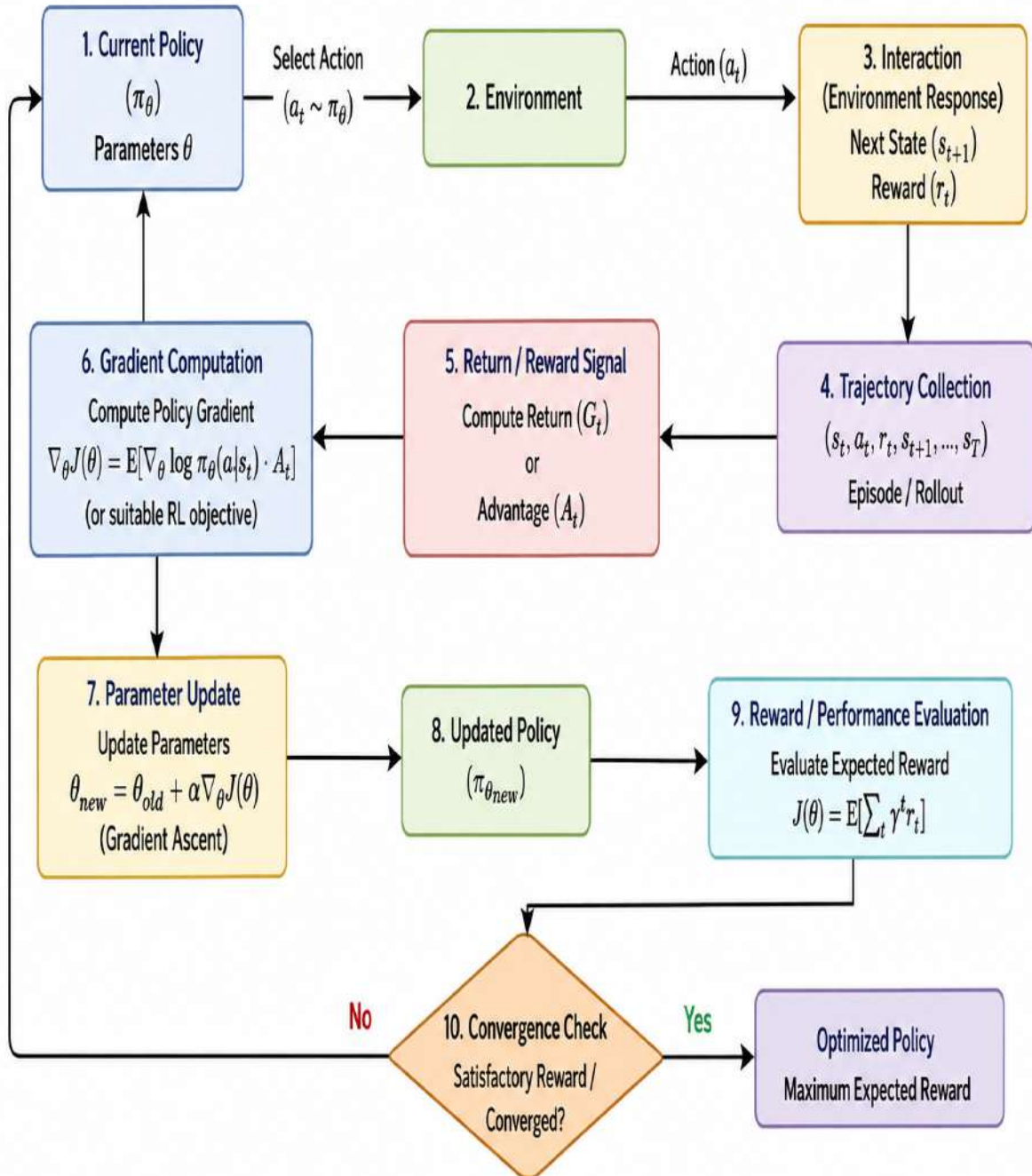
Simple Explanation

1. **Policy Parameters:** Define the current policy.
2. **Actions:** Policy produces actions.
3. **Environment:** Gives feedback.
4. **Rewards:** Measure performance.
5. **Gradient:** Calculates improvement direction.
6. **Update:** Parameters are changed.
7. **Optimization:** Higher rewards are targeted.

Key Idea

Compute Gradient → Update Parameters → Maximize Reward

Gradient Computation, Parameter Updates and Reward Optimization in Reinforcement Learning



9. Gradient-Based Optimization Using Backpropagation

Question: Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.

Flowchart

Start → Initialize Neural Network → Input State → Select Action → Receive Reward → Calculate Loss → Backpropagation → Compute Gradients → Update Weights → Improved Policy

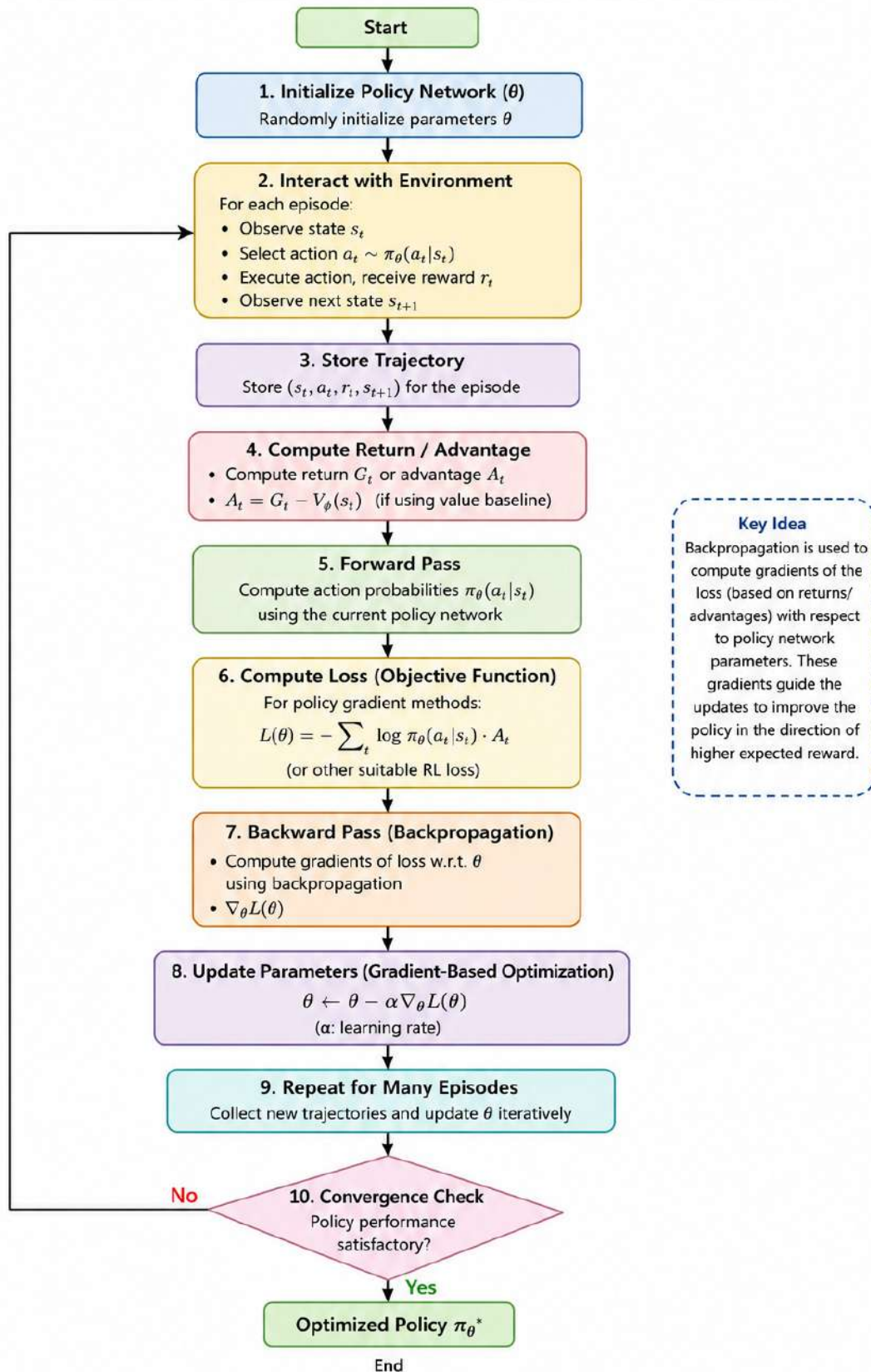
Simple Explanation

1. **Network:** Represents the policy.
2. **Input:** Current state is given to the network.
3. **Action:** Network selects an action.
4. **Reward:** Environment gives feedback.
5. **Loss:** Performance difference is calculated.
6. **Backpropagation:** Gradients are calculated.
7. **Update:** Network weights are changed.

Key Idea

Backpropagation calculates gradients and updates network weights to improve the policy.

Gradient-Based Optimization Using Backpropagation in Reinforcement Learning



10. Policy Parameters, Gradients, Optimization Algorithms and Rewards

Question: Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.

Concept Map

**Policy Parameters → Policy → Actions → Rewards → Objective →
Gradients → Optimization Algorithm → Updated Parameters**

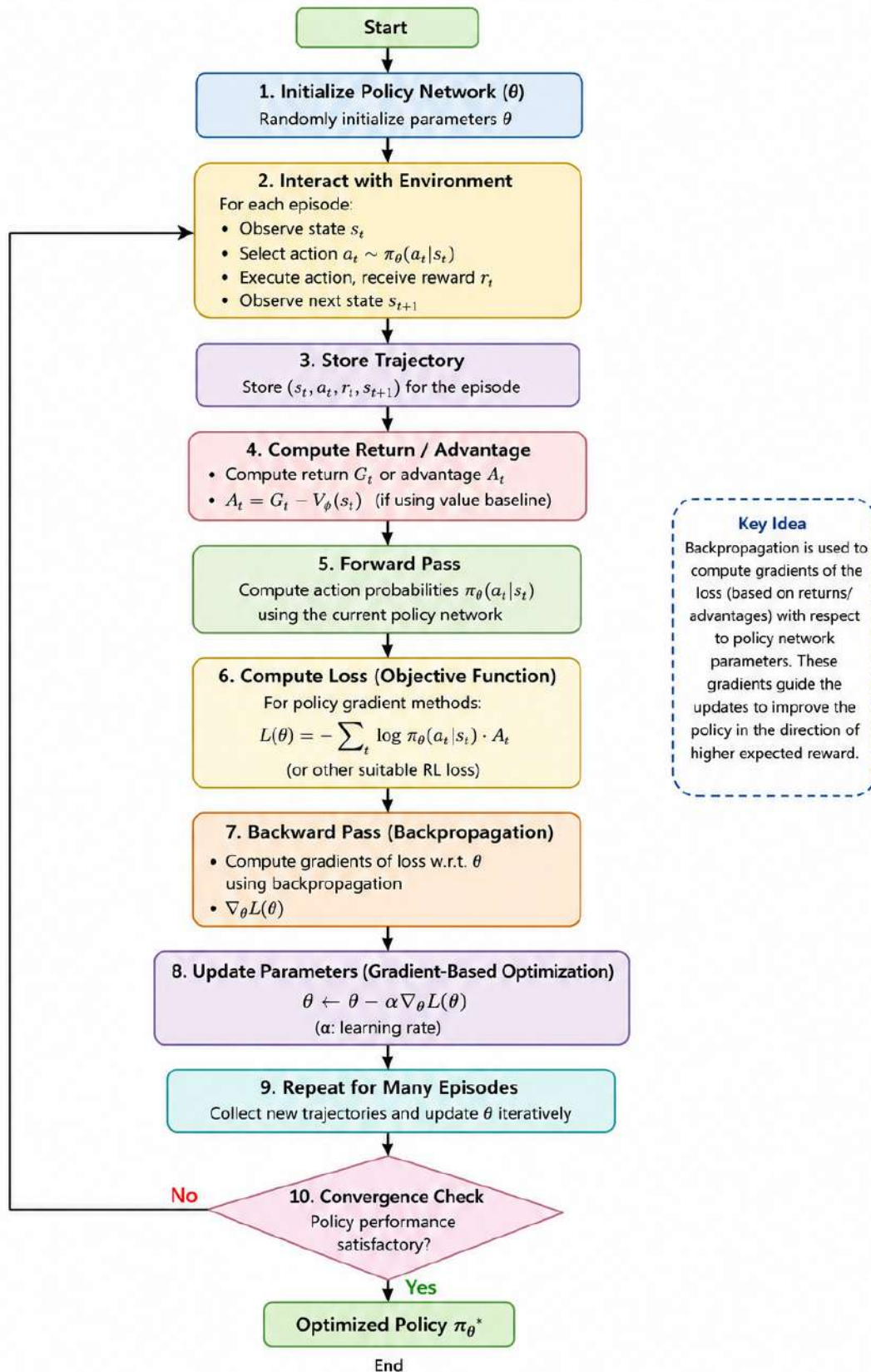
Simple Explanation

1. **Policy Parameters:** Control the policy.
2. **Actions:** Are selected by the policy.
3. **Rewards:** Measure performance.
4. **Gradients:** Show the improvement direction.
5. **Optimization Algorithm:** Updates parameters.
6. **Updated Parameters:** Produce a better policy.

Key Idea

Rewards guide gradients, and optimization algorithms use gradients to improve parameters.

Gradient-Based Optimization Using Backpropagation in Reinforcement Learning



Sampling-Based Planning

11. Sampling-Based Planning Process

Question: Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

Flowchart

Start → Learn Environment Model → Observe State → Sample Actions → Generate Trajectories → Calculate Rewards → Evaluate Trajectories → Select Best Action → Execute Action

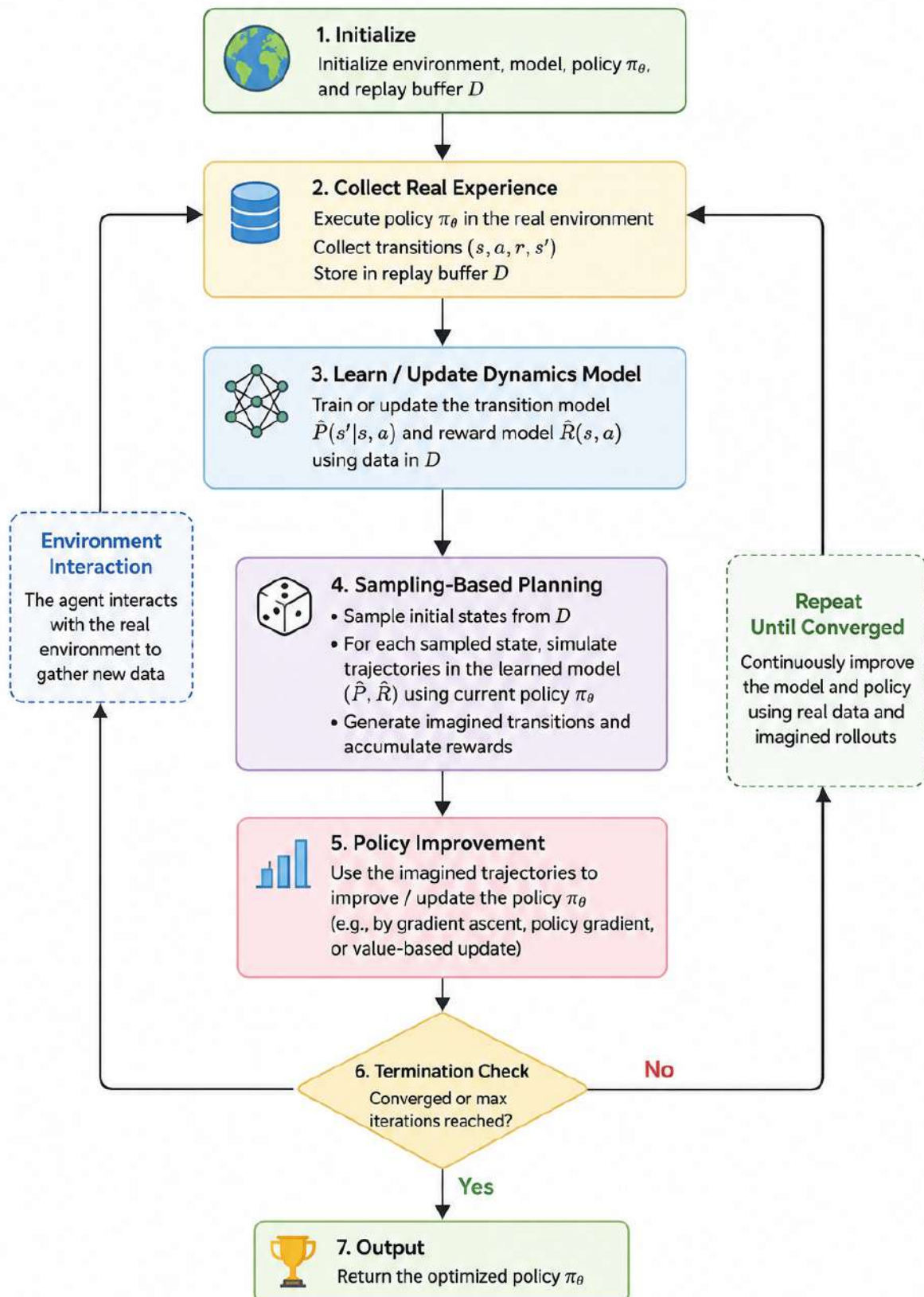
Simple Explanation

1. **Model:** Provides predicted environment behavior.
2. **State:** Current state is observed.
3. **Sampling:** Possible actions are sampled.
4. **Trajectories:** Future paths are generated.
5. **Evaluation:** Rewards are calculated.
6. **Selection:** Best action is chosen.

Key Idea

Sample possible futures → Evaluate them → Choose the best action.

Sampling-Based Planning Process in Model-Based Reinforcement Learning



12. Sampled Trajectories for Planning

Question: Prepare a concept map showing how sampled trajectories are generated and evaluated for planning.

Concept Map

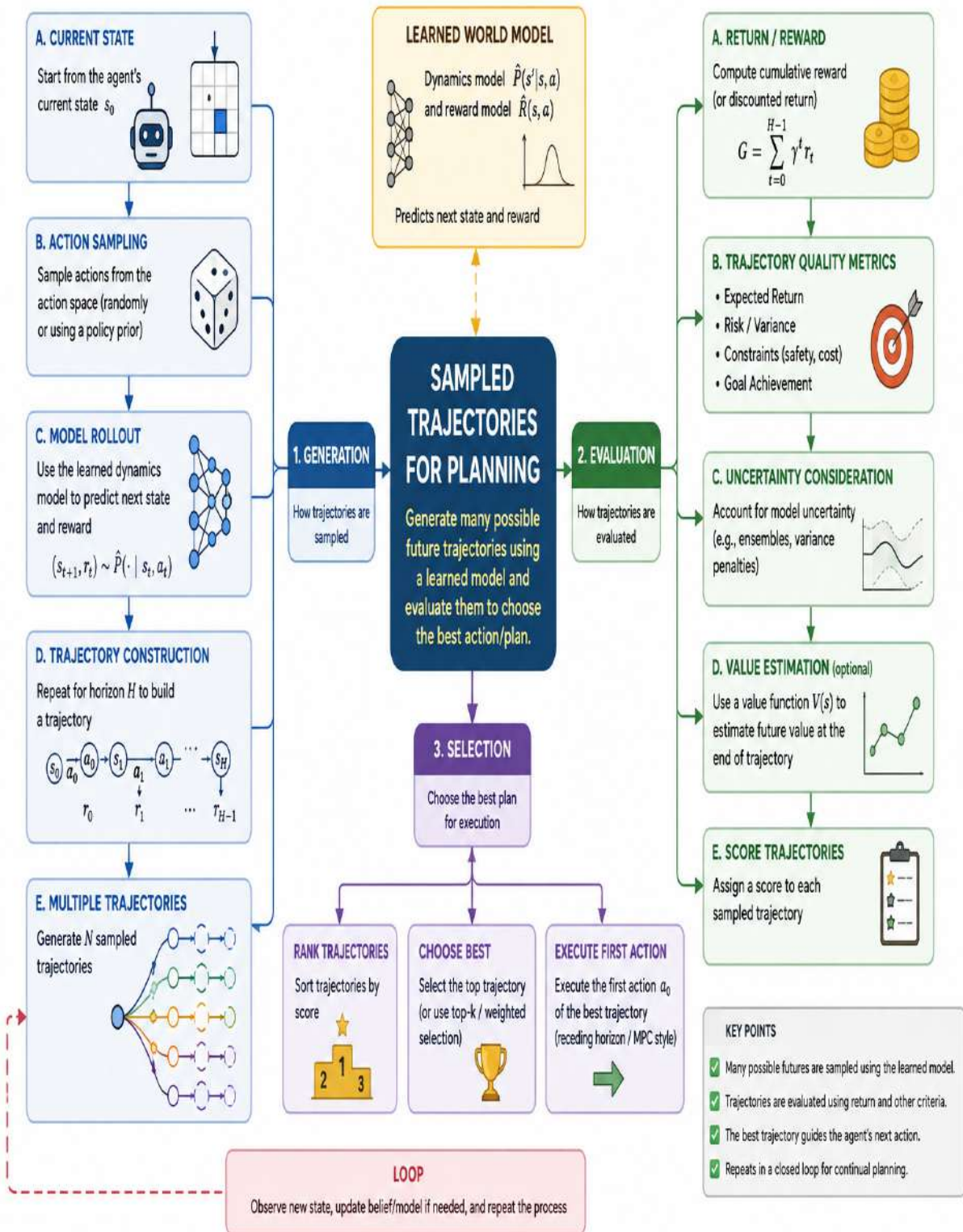
Environment Model → Current State → Sample Actions → Generate Trajectories → Calculate Returns → Evaluate → Select Best Trajectory → Best Action

Simple Explanation

1. **Model:** Predicts future behavior.
2. **Actions:** Several actions are sampled.
3. **Trajectories:** Possible future paths are generated.
4. **Returns:** Rewards are calculated.
5. **Evaluation:** Trajectories are compared.
6. **Best:** Highest-value trajectory is selected.

Key Idea

Generate → Evaluate → Compare → Select



13. Environment Model, Planner, Sampled Actions and Policy

Question: Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy.

Block Diagram

Environment Model → Planner → Sampled Actions → Action Evaluation → Best Action → Optimal Policy

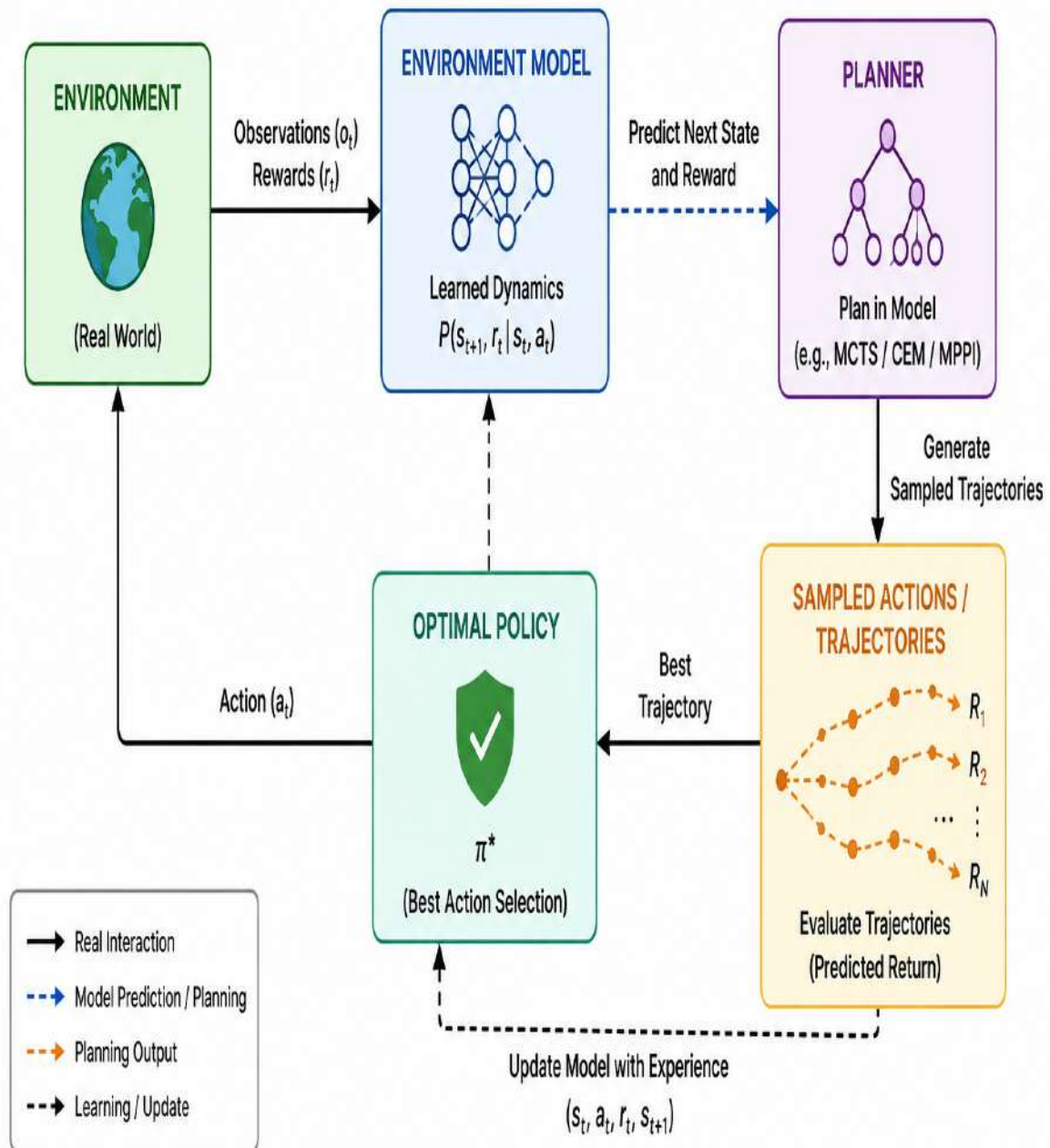
Simple Explanation

1. **Environment Model:** Predicts future states.
2. **Planner:** Uses the model.
3. **Sampled Actions:** Different actions are tested.
4. **Evaluation:** Their expected rewards are calculated.
5. **Optimal Policy:** The best action is selected.

Key Idea

The planner uses the model to test actions and find a good policy.

Environment Model, Planner, Sampled Actions and Policy



14. Monte Carlo Sampling for Planning

Question: Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.

Flowchart

Start → Select State → Randomly Sample Action → Generate Trajectory → Calculate Return → Repeat Simulations → Average Returns → Compare Actions → Select Best Action

Simple Explanation

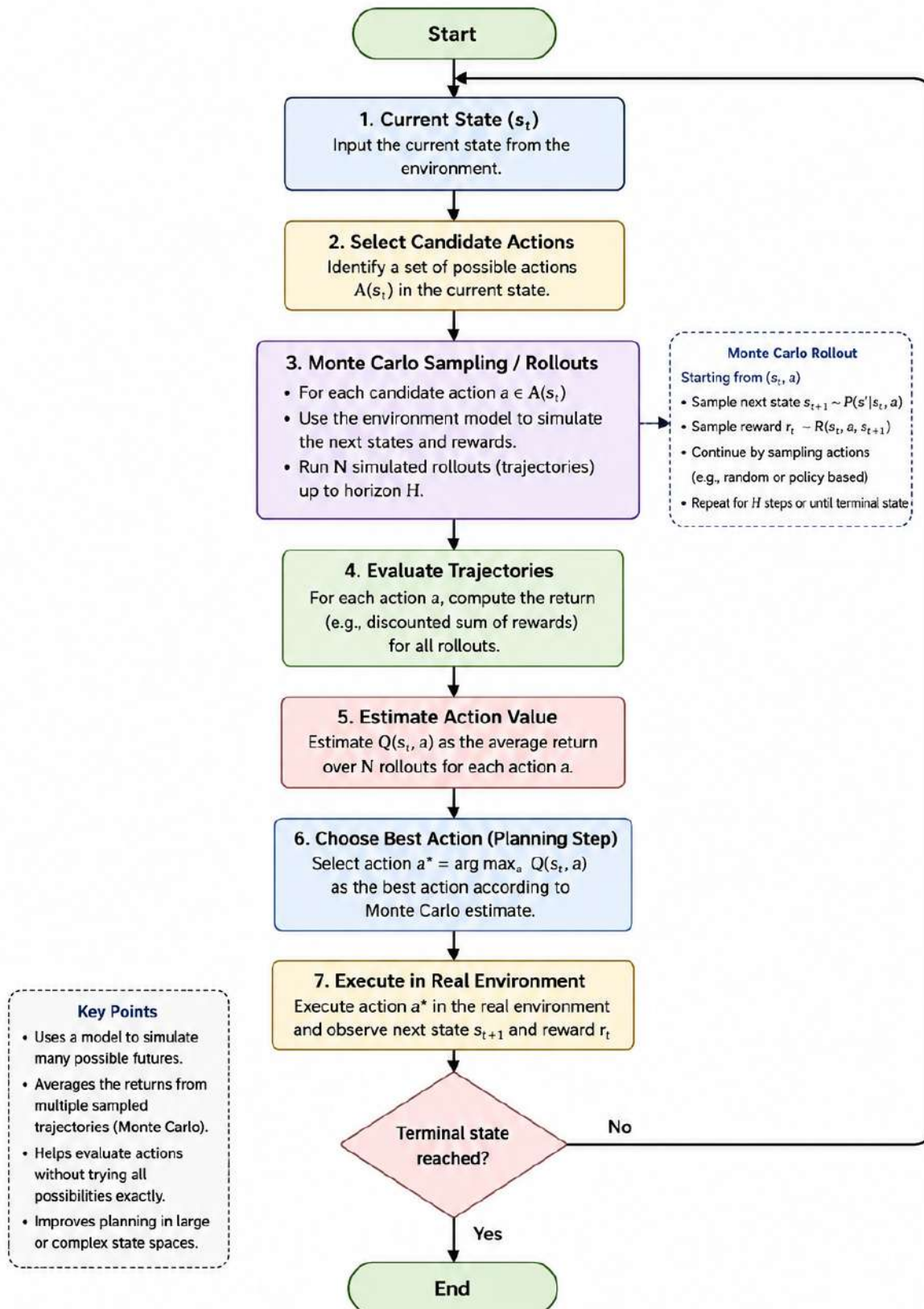
1. **Select State:** Start from the current state.
2. **Sampling:** Random actions are selected.
3. **Simulation:** Future trajectories are generated.
4. **Return:** Total reward is calculated.
5. **Repeat:** Many simulations are performed.
6. **Average:** Returns are averaged.
7. **Select:** Action with the best expected return is chosen.

Key Idea

Monte Carlo planning uses repeated simulations to estimate the best action.

MONTE CARLO SAMPLING FOR PLANNING

(Model-Based Reinforcement Learning)



15. Exhaustive Search vs Sampling-Based Planning

Question: Draw a concept map comparing exhaustive search planning and sampling-based planning techniques.

Concept Map

Planning → Exhaustive Search / Sampling-Based Planning

Exhaustive Search → Many possibilities → High computation → More complete

Sampling-Based → Selected possibilities → Lower computation → Approximate

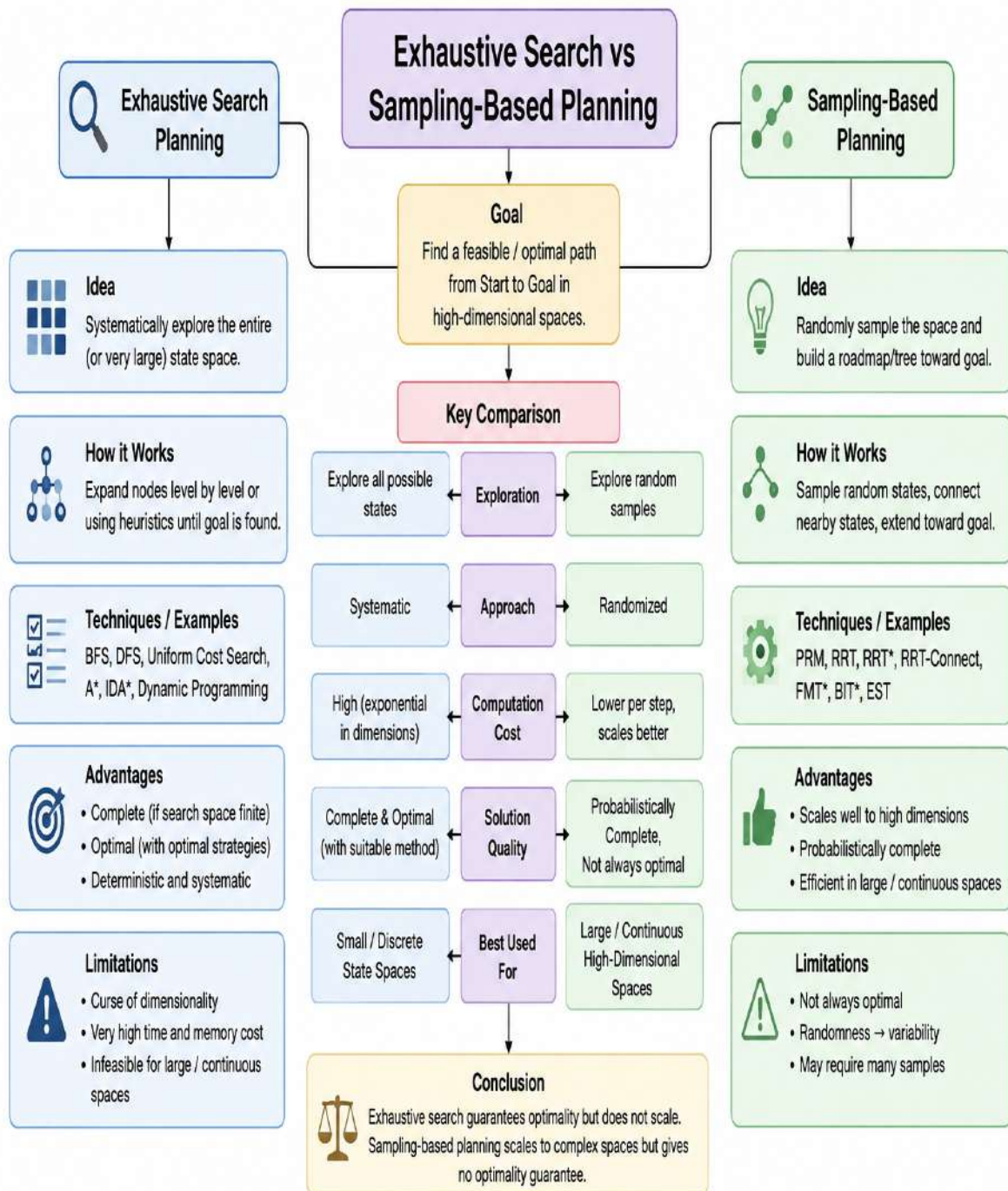
Simple Explanation

1. **Exhaustive Search:** Examines many or all possible paths.
2. **Advantage:** Can provide more complete results.
3. **Disadvantage:** Requires high computation.
4. **Sampling-Based:** Examines selected possible paths.
5. **Advantage:** Faster and less computationally expensive.
6. **Disadvantage:** Results are approximate.

Key Idea

Exhaustive = More computation

Sampling = Faster approximate planning



Model-Based Data Generation

16. Synthetic Data Generation Using Learned Model

Question: Draw a flowchart illustrating synthetic data generation using a learned environment model.

Flowchart

Start → Collect Real Experience → Learn Environment Model → Give State & Action → Predict Next State & Reward → Generate Synthetic Experience → Store Data → Train Policy → Improved Policy

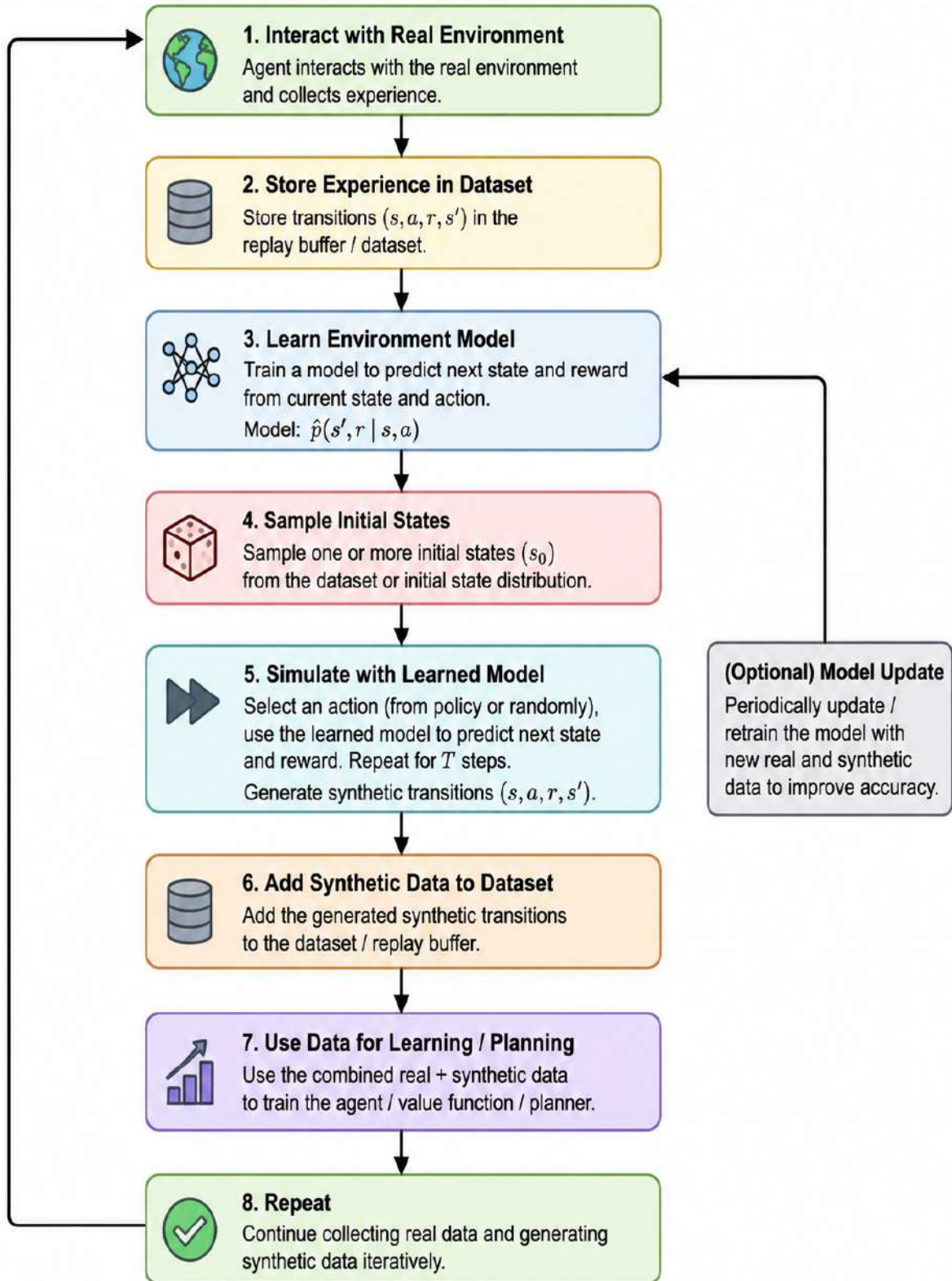
Simple Explanation

1. **Real Experience:** Data is collected from the environment.
2. **Learn Model:** Environment behavior is learned.
3. **Prediction:** Next state and reward are predicted.
4. **Synthetic Data:** Artificial experiences are generated.
5. **Training:** Synthetic data is used to train the policy.
6. **Improvement:** Policy performance increases.

Key Idea

Learn Model → Generate Synthetic Data → Train Policy

Synthetic Data Generation Using Learned Model



17. Environment Model, Generated Experiences, Replay Buffer and Policy

Question: Prepare a concept map showing the relationship among environment model, generated experiences, replay buffer, and policy learning.

Concept Map

Environment → Real Experience → Environment Model → Generated Experience → Replay Buffer → Policy Learning → Improved Policy

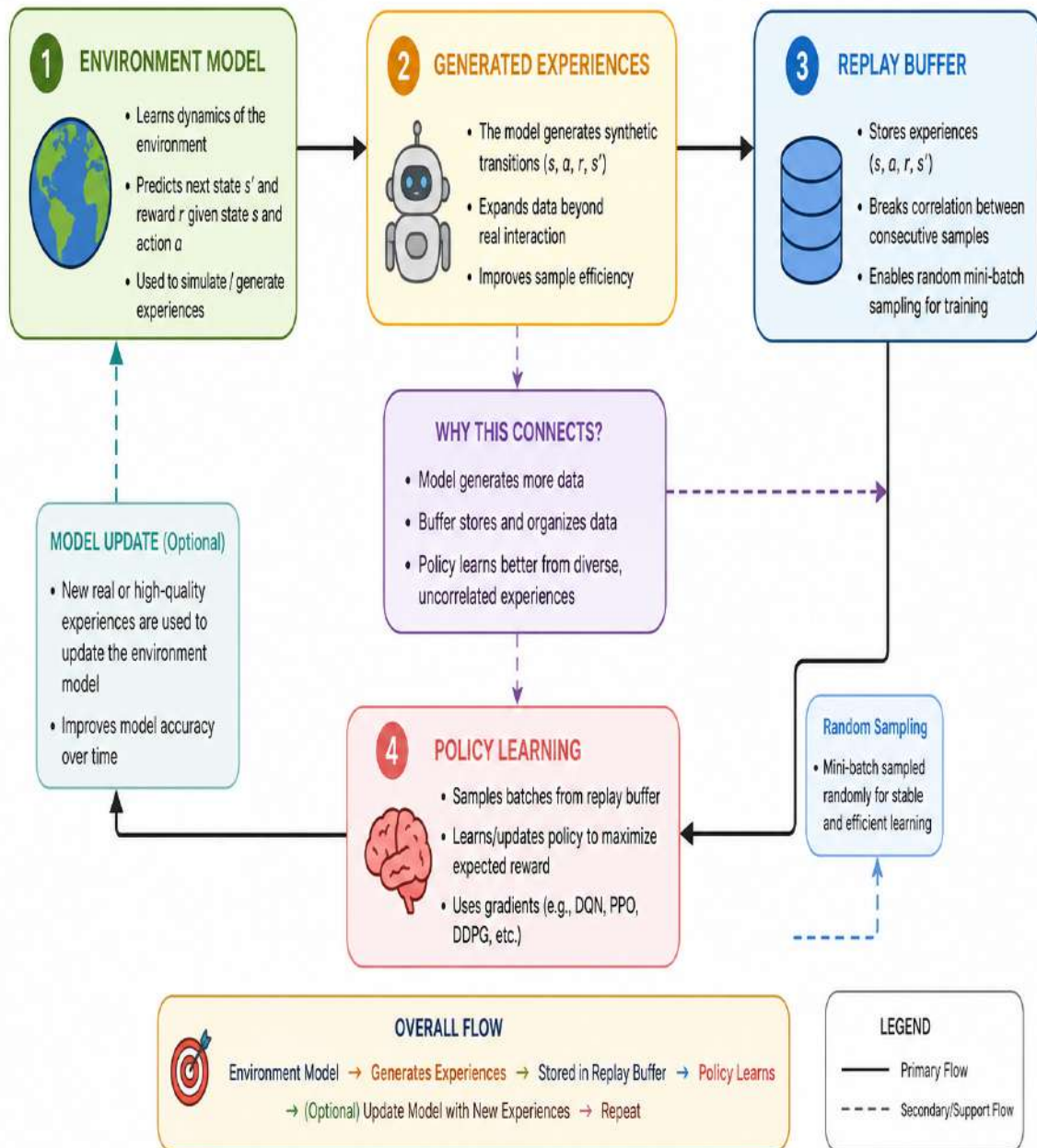
Simple Explanation

1. **Environment:** Provides real experiences.
2. **Model:** Learns environment behavior.
3. **Generated Experience:** Model creates additional data.
4. **Replay Buffer:** Stores experiences.
5. **Policy Learning:** Uses stored data for training.
6. **Improved Policy:** Learns better actions.

Key Idea

Model-generated experiences provide extra data for policy learning.

Relationship Between Environment Model, Generated Experiences, Replay Buffer and Policy



18. Model-Generated Experience Replay

Question: Draw a block diagram explaining model-generated experience replay in Reinforcement Learning.

Block Diagram

Real Environment → Learn Model → Generate Experiences → Replay Buffer → Policy Training → Improved Policy

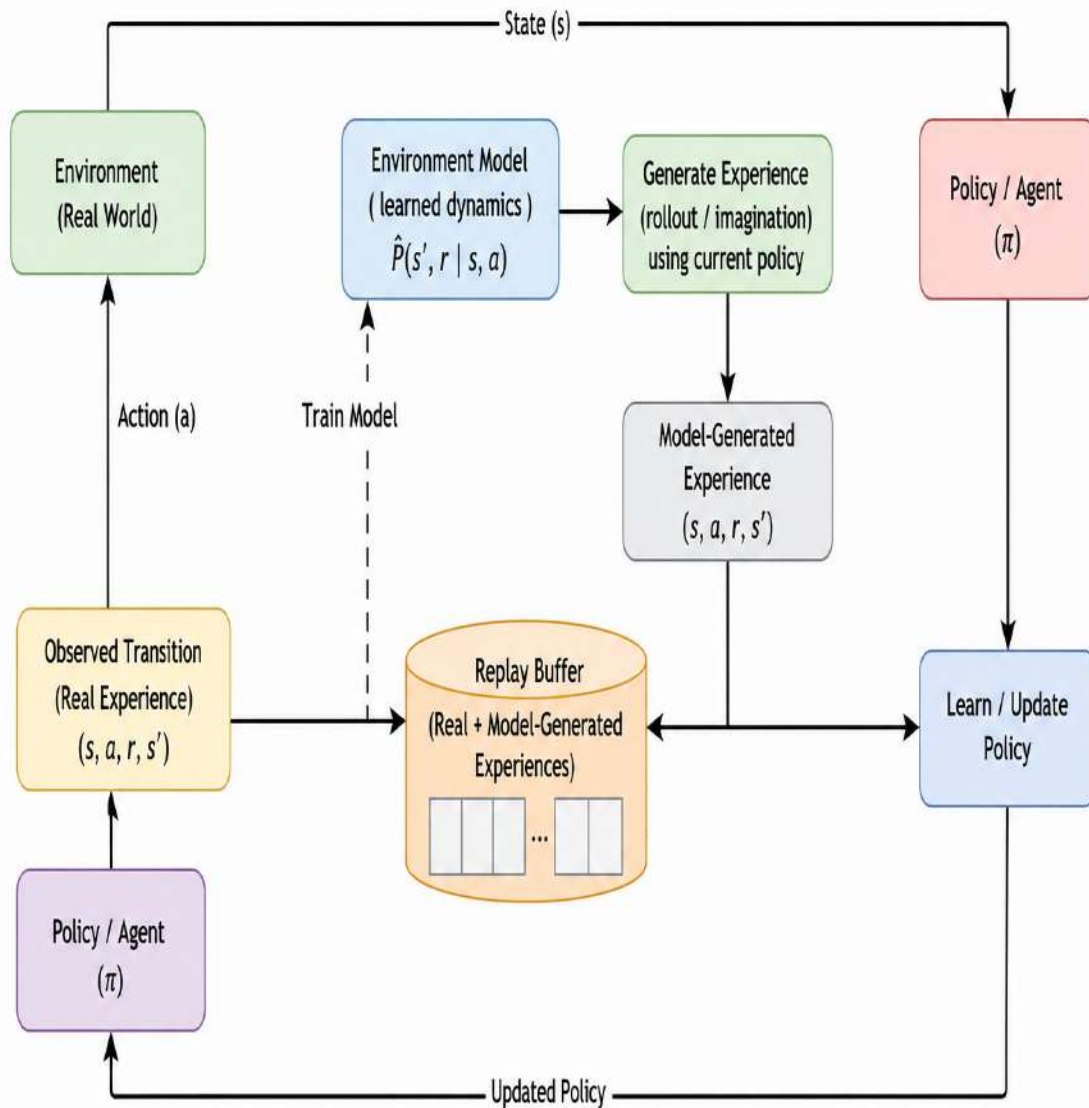
Simple Explanation

1. **Environment:** Provides real data.
2. **Model:** Learns environment dynamics.
3. **Generated Experiences:** Model creates additional experiences.
4. **Replay Buffer:** Stores real and generated experiences.
5. **Training:** Agent learns from stored experiences.
6. **Policy:** Becomes better.

Key Idea

Model-generated data increases the amount of experience available for training.

MODEL-GENERATED EXPERIENCE REPLAY



Flow Summary

Real Experience $\rightarrow$ Train Model $\rightarrow$ Generate Imagined Experience $\rightarrow$
Store in Replay Buffer $\rightarrow$ Sample Batch $\rightarrow$ Learn / Update Policy

19. Simulated Experiences and Sample Efficiency

Question: Design a flowchart showing how simulated experiences improve sample efficiency during training.

Flowchart

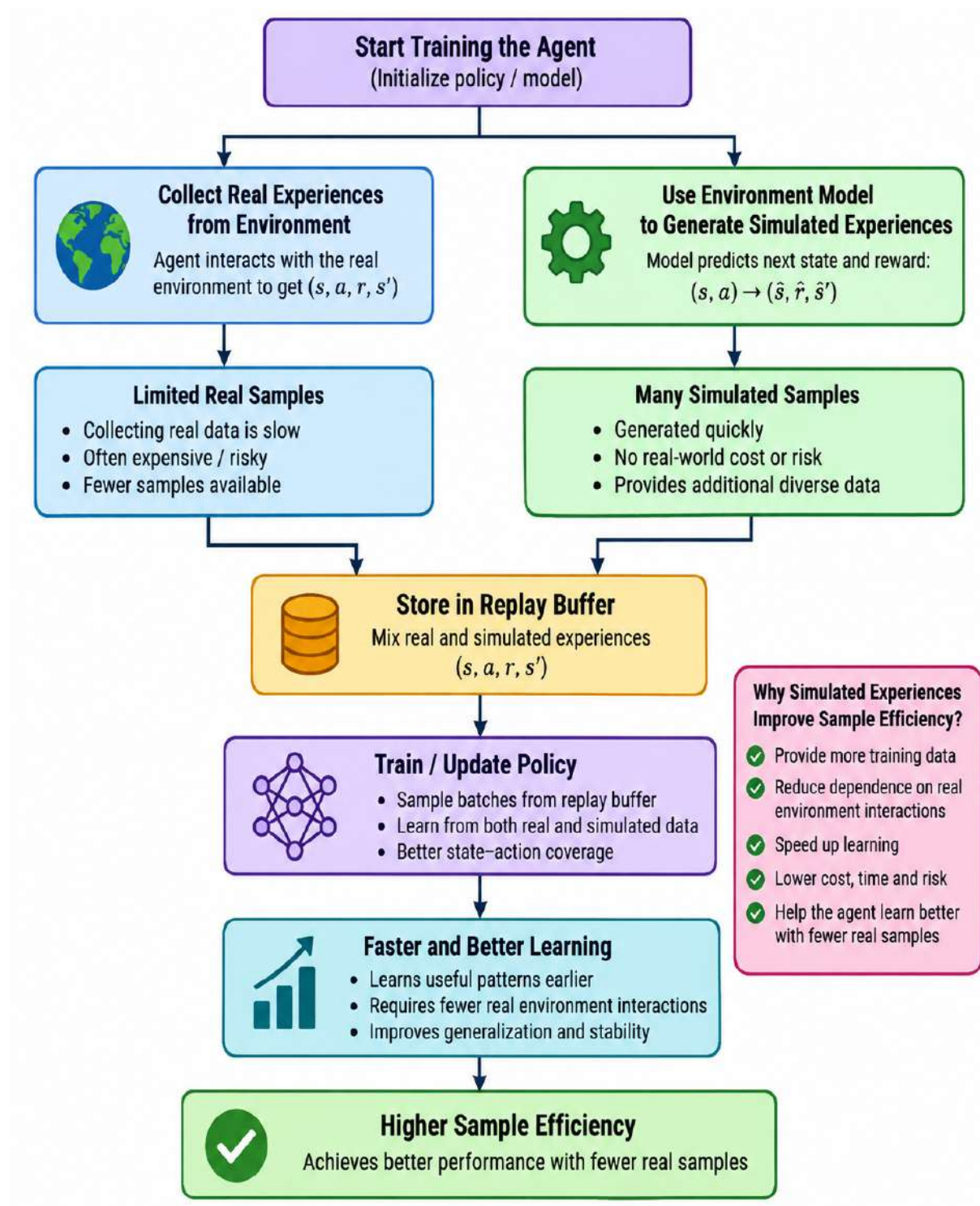
Real Environment → Collect Limited Data → Learn Model → Generate Simulated Experiences → Combine Real + Simulated Data → Train Agent → Better Sample Efficiency → Improved Policy

Simple Explanation

1. **Real Data:** A small amount of real experience is collected.
2. **Learn Model:** The environment model is trained.
3. **Simulation:** More experiences are generated.
4. **Combine:** Real and simulated data are used.
5. **Training:** Agent learns from more data.
6. **Efficiency:** Fewer real interactions are needed.

Key Idea

More simulated data → Less real-world interaction → Better sample efficiency.



20. Advantages and Limitations of Model-Generated Training Data

Question: Draw a concept map illustrating the advantages and limitations of model-generated training data.

Concept Map

Model-Generated Data → Advantages / Limitations

Advantages → More Data → Faster Training → Lower Cost → Better Sample Efficiency

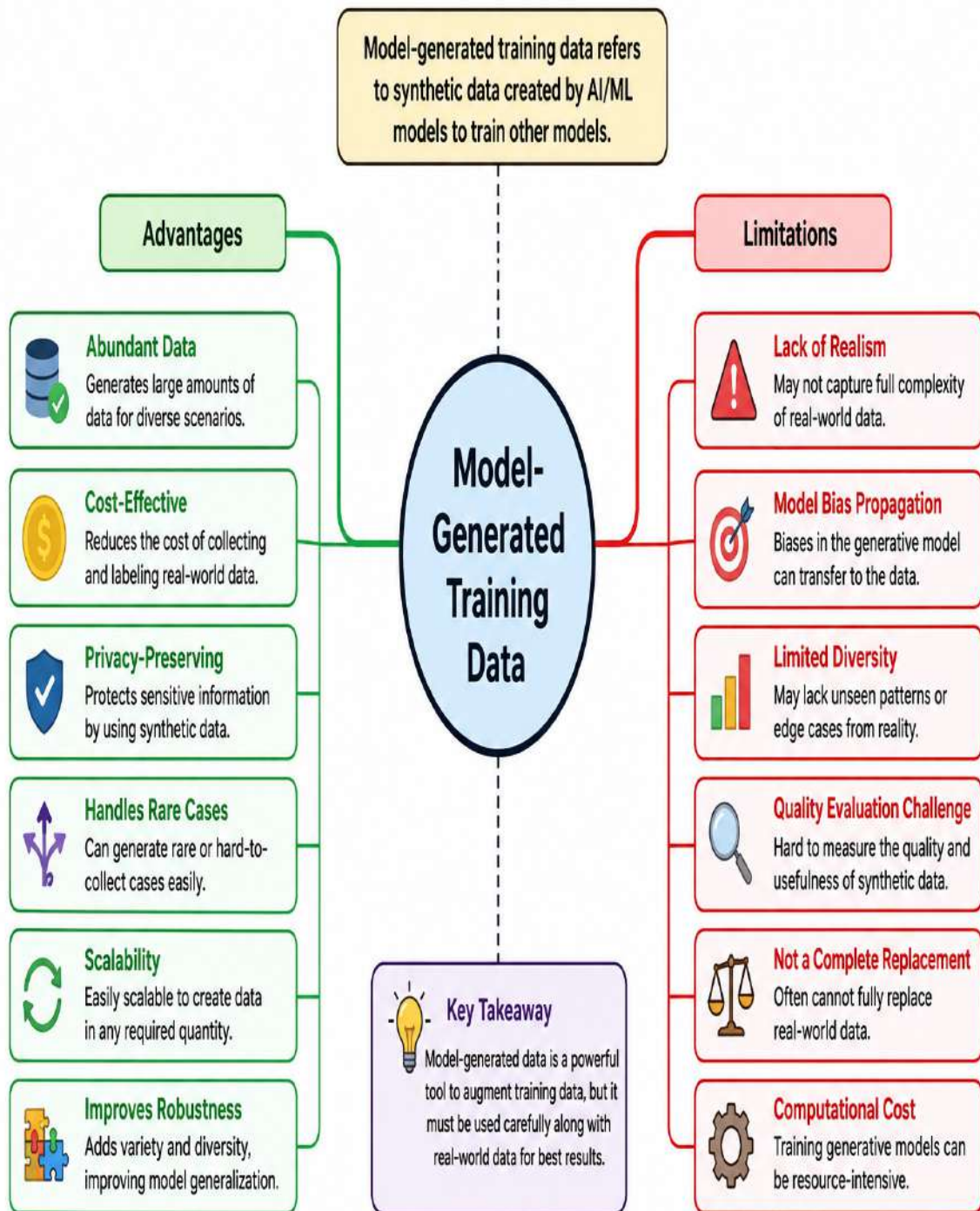
Limitations → Model Errors → Inaccurate Data → Bias → Poor Policy

Simple Explanation

1. **More Data:** Provides additional training experiences.
2. **Lower Cost:** Reduces real-world interaction.
3. **Faster Training:** More data can speed learning.
4. **Model Errors:** Incorrect models create incorrect data.
5. **Bias:** Generated data may contain model bias.
6. **Uncertainty:** Predictions may not always be accurate.

Key Idea

Useful synthetic data can improve learning, but incorrect models can reduce performance.



Value-Equivalence Prediction and Policy Optimization

21. Value-Equivalence Prediction

Question: Draw a flowchart explaining value-equivalence prediction for estimating future rewards.

Flowchart

Start → Observe State → Select Action → Use Learned Model → Predict Future States → Predict Rewards → Estimate Value → Compare Actions → Select High-Value Action

Simple Explanation

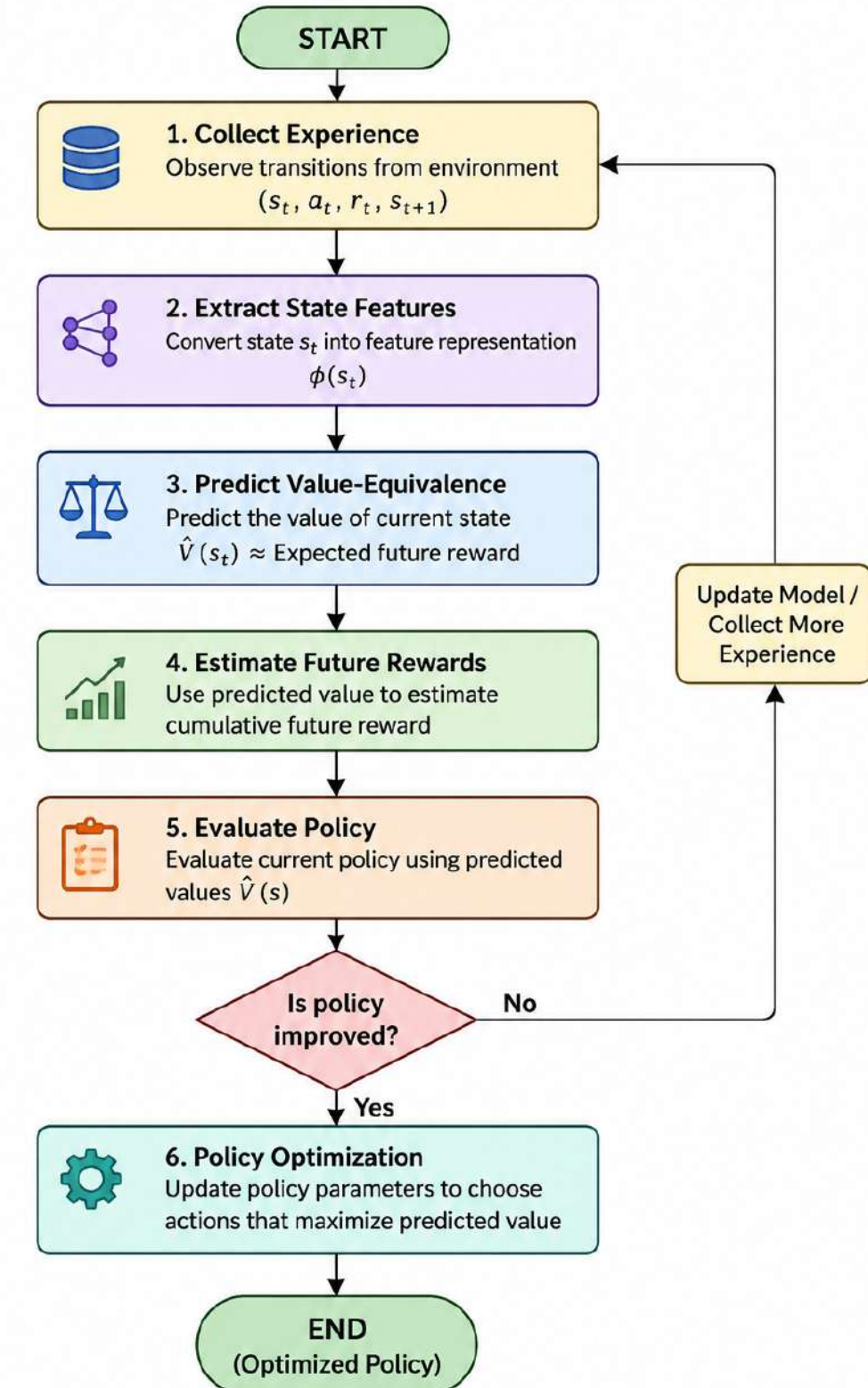
1. **State:** Current state is observed.
2. **Action:** Possible actions are considered.
3. **Model:** Predicts future states.
4. **Reward:** Future rewards are predicted.
5. **Value:** Expected future reward is estimated.
6. **Comparison:** Actions are compared.
7. **Selection:** High-value action is selected.

Key Idea

Value prediction estimates future rewards to help select better actions.

VALUE-EQUIVALENCE PREDICTION

Estimating Future Rewards



22. Value Functions, Reward Prediction, Policy Evaluation and Improvement

Question: Prepare a concept map showing the relationship among value functions, reward prediction, policy evaluation, and policy improvement.

Concept Map

Reward Prediction → Future Rewards → Value Function → Policy Evaluation → Policy Improvement → Better Policy → Higher Rewards

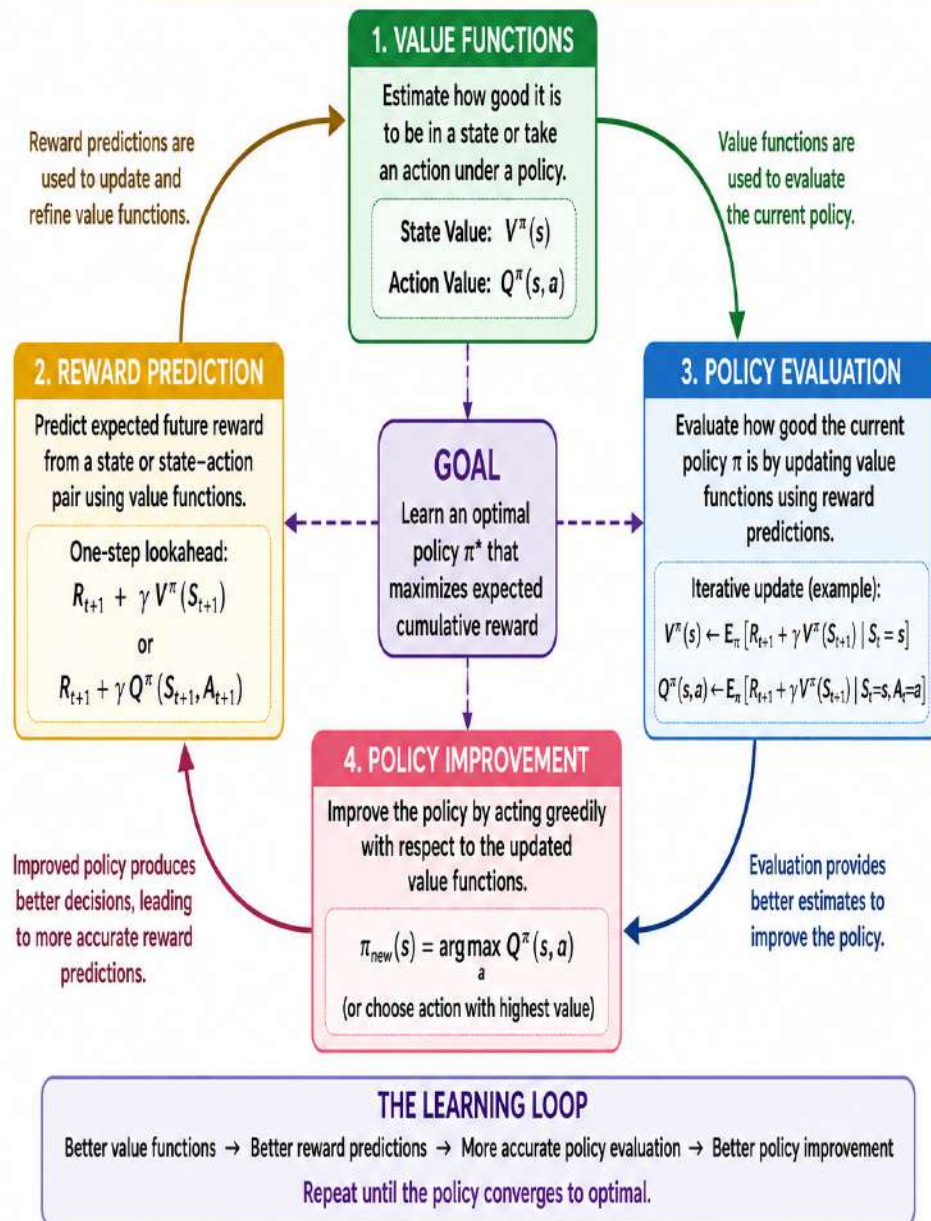
Simple Explanation

1. **Reward Prediction:** Estimates future rewards.
2. **Value Function:** Represents expected future returns.
3. **Policy Evaluation:** Measures policy quality.
4. **Policy Improvement:** Changes the policy for better performance.
5. **Higher Rewards:** Improved policy produces better results.

Key Idea

Predict rewards → Estimate value → Evaluate policy → Improve policy.

VALUE FUNCTIONS, REWARD PREDICTION, POLICY EVALUATION & POLICY IMPROVEMENT



23. Model-Based Policy Optimization (MBPO)

Question: Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.

Block Diagram

Real Environment → Collect Data → Learn Dynamics Model → Generate Short Model Rollouts → Synthetic Experiences → Policy Optimization → Improved Policy → Environment

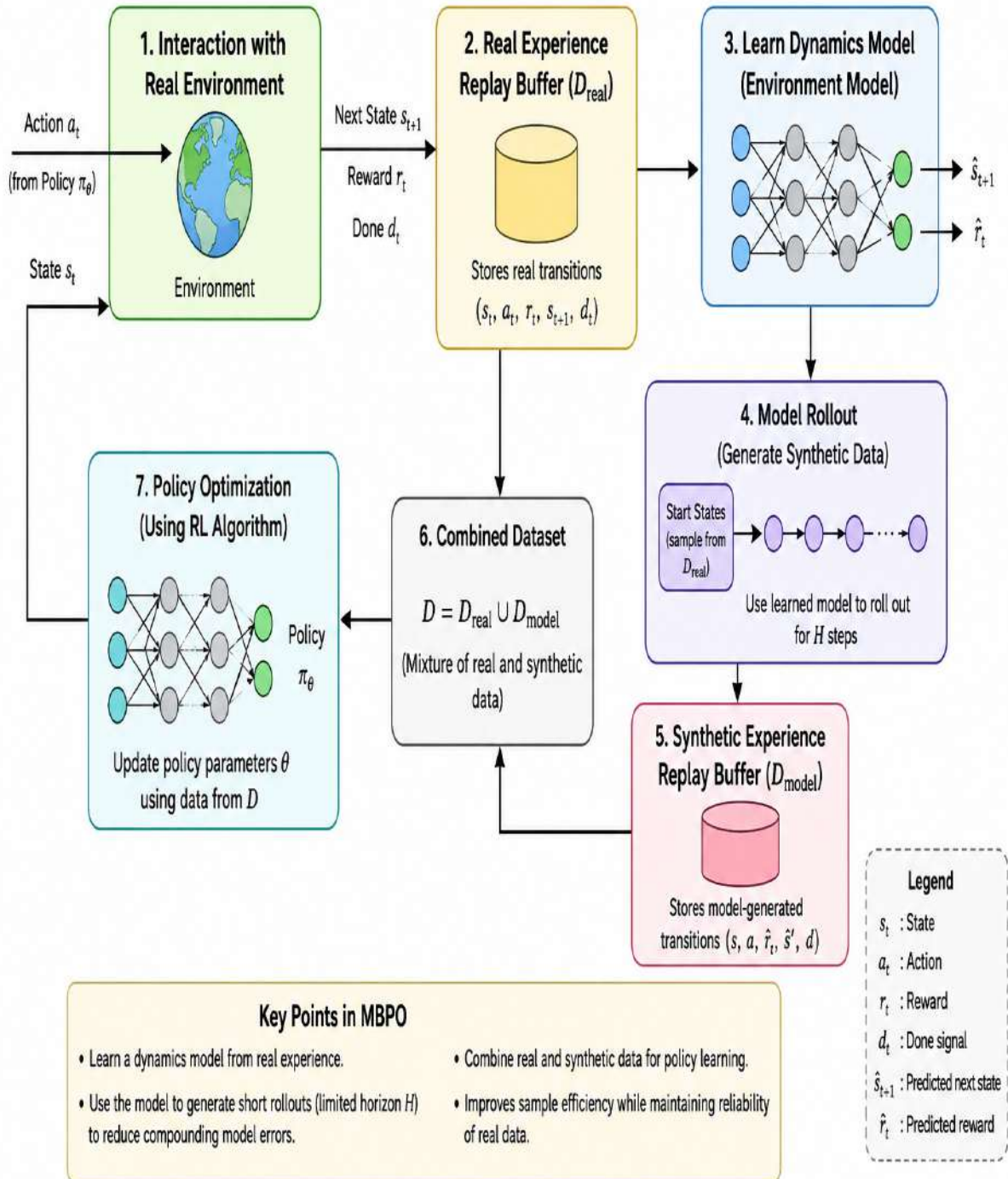
Simple Explanation

1. **Environment:** Provides real experiences.
2. **Data Collection:** Real data is collected.
3. **Dynamics Model:** Learns environment behavior.
4. **Model Rollouts:** Short simulated experiences are generated.
5. **Synthetic Data:** Additional training data is created.
6. **Policy Optimization:** Policy is improved.
7. **Environment:** Improved policy interacts again.

Key Idea

MBPO combines real experience with short model-generated rollouts to improve policy learning.

Model-Based Policy Optimization (MBPO) Framework



24. Iterative Policy Optimization Using Learned Dynamics

Question: Design a flowchart explaining iterative policy optimization using learned environment dynamics.

Flowchart

Start → Collect Real Experience → Learn Environment Dynamics → Generate Predictions → Evaluate Policy → Optimize Policy → Execute Improved Policy → Collect New Experience → Update Model → Repeat

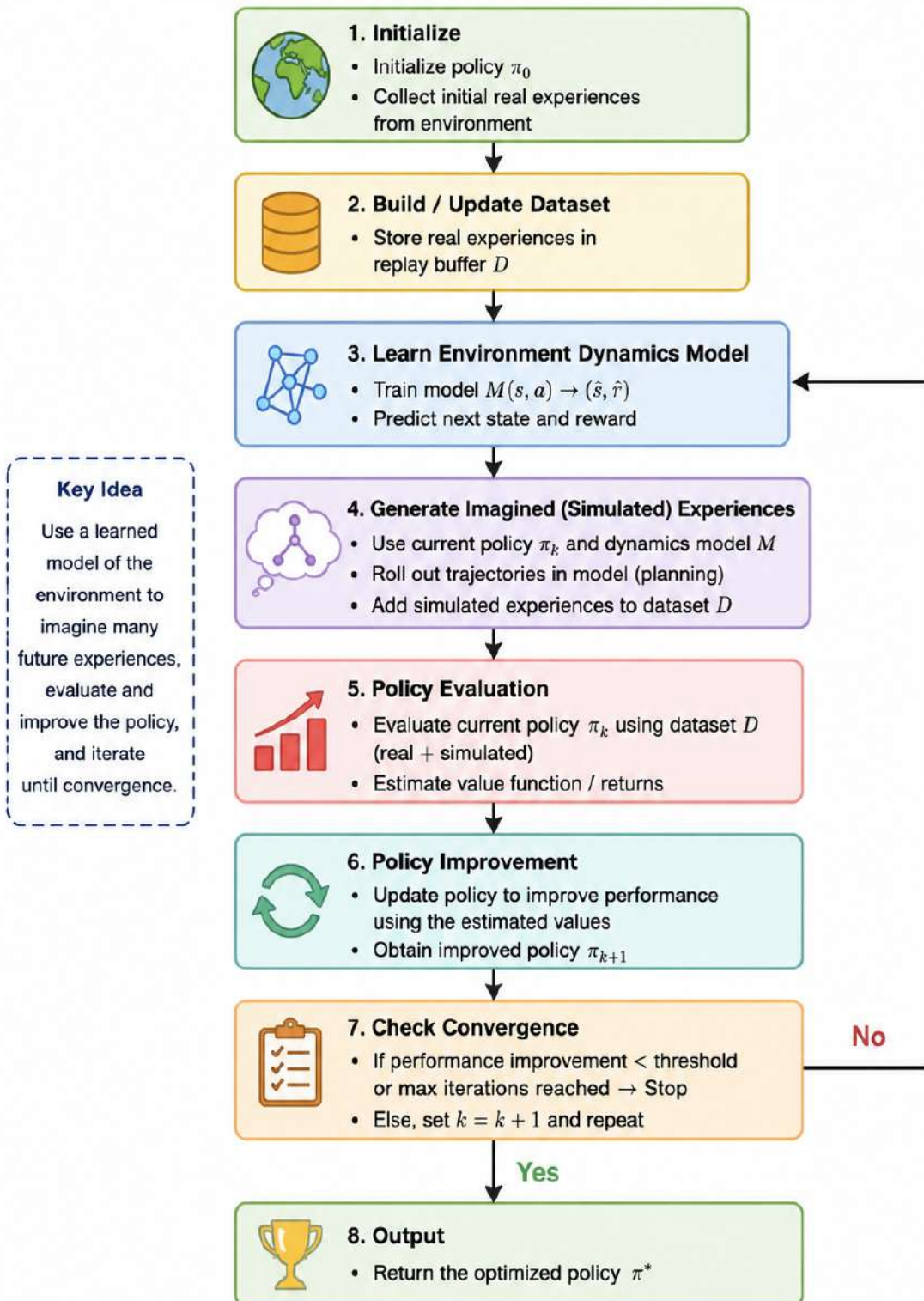
Simple Explanation

1. **Experience:** Collect data from the environment.
2. **Learn Dynamics:** Build an environment model.
3. **Prediction:** Predict future outcomes.
4. **Evaluation:** Evaluate current policy.
5. **Optimization:** Improve the policy.
6. **New Experience:** Use the improved policy.
7. **Repeat:** Update the model and policy again.

Key Idea

Learn → Predict → Evaluate → Optimize → Execute → Repeat

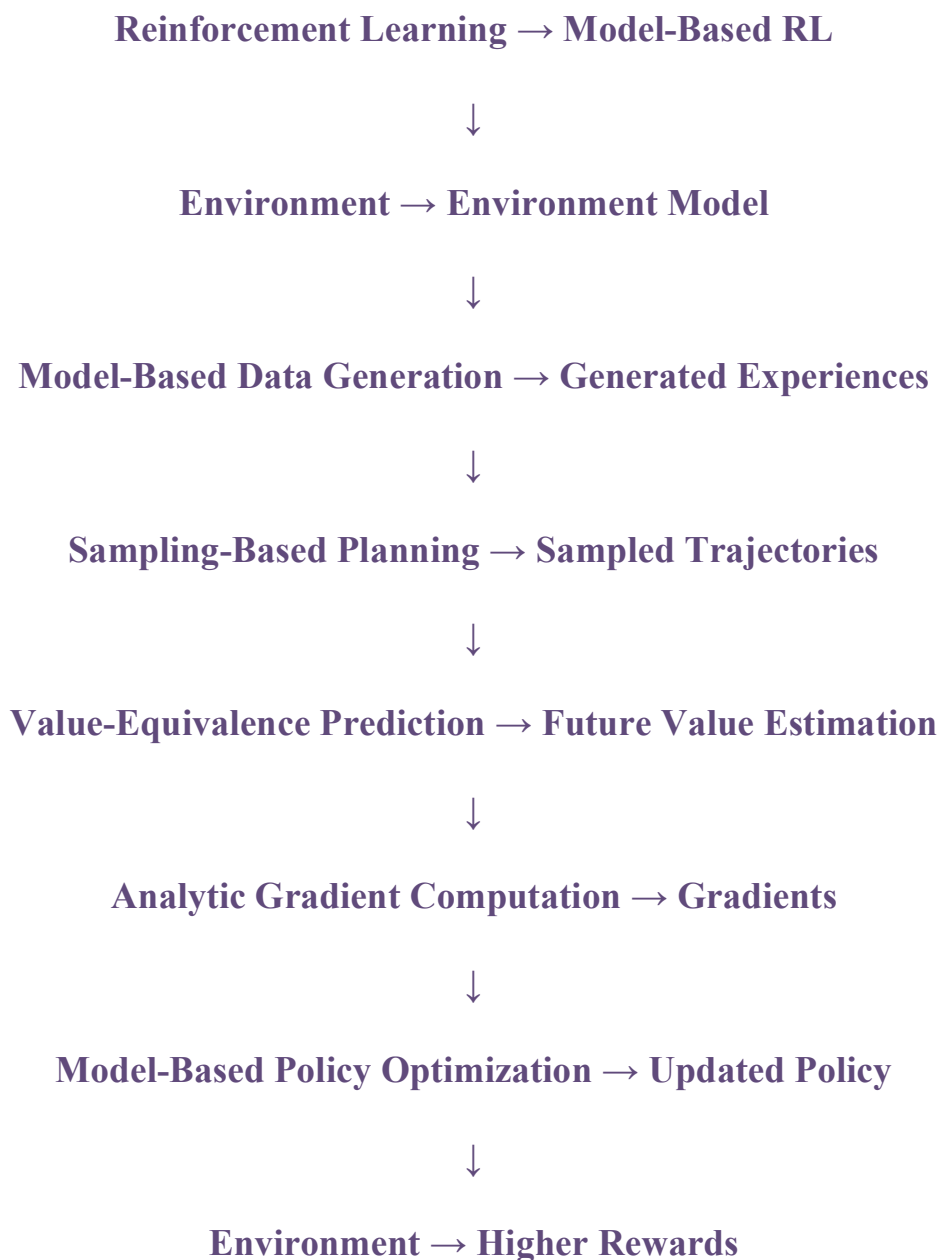
Iterative Policy Optimization Using Learned Environment Dynamics



25. Comprehensive Model-Based RL Framework

Question: Draw a comprehensive concept map integrating Model-Based RL, Analytic Gradient Computation, Sampling-Based Planning, Model-Based Data Generation, Value-Equivalence Prediction, and Model-Based Policy Optimization into a unified Reinforcement Learning framework.

Concept Map



Simple Explanation

1. **Model-Based RL:** Learns how the environment behaves.
2. **Data Generation:** Creates additional simulated experiences.
3. **Sampling-Based Planning:** Tests possible future trajectories.
4. **Value Prediction:** Estimates future rewards.
5. **Gradient Computation:** Finds how the policy should change.
6. **Policy Optimization:** Updates the policy.
7. **Environment:** The improved policy is executed.
8. **Repeat:** New experience is used to improve the system again.

Key Idea

Model-Based RL → Learn Model → Generate Data → Plan → Predict Values → Compute Gradients → Optimize Policy → Higher Rewards

